Anomaly Signature Bench (ASB) v2

Head-mounted multi-sensor rig and Android operator app: v1 audit, 15 new modules, settled defaults

Supersedes *Glasses-Detection-Build-Spec-v1*. Specification only: nothing has been bought, built, sent, installed or published, and nothing touches other projects. Prices and facts were checked on **2026-10-28** (search listings unless noted). Unverified items are tagged **[UNVERIFIED]**; autonomous choices are tagged **[DEFAULT]** and listed in Section 15. Per the owner's instruction, every decision is settled as a recorded default; only a purchase approval and a few facts about the owner's own hardware remain open.

Summary: what v2 decides

Question	Answer in this spec
What changed?	The v1 audit was repeated and found **32 more gaps** (Section 1; all 30 v1 rows kept). Fifteen modules were added (Section 5): eye behavior, liveness stacking, signal fingerprints, security, sighting log, voice output, location baselines, thermal grayscale, headless test + golden dataset, macOS build, sensor adapter layer, feedback loop, Android fallback node, sweep timeline, app gate. Architecture is unchanged: **one process, a thread per sensor, scores in 0..1, a fusion layer**.
Most important honest findings	(1) **Micro-saccades cannot be measured with this rig at any useful distance**: they need ≥250 Hz sampling and about 0.1 mm of pupil shift, which is under one pixel at 0.5 m. Pupil size, blink and corneal-glint behavior at ≤1 m are feasible; they are research signals, not detectors. (2) **A silicone mask worn by a live person keeps the wearer's real eyes**, so eye signals cannot catch it; thermal and skin-surface cues may. (3) The RTL-SDR V4 tunes only **500 kHz–1.766 GHz** and cannot see 2.4 GHz BLE/Wi-Fi at all, so IQ-level radio fingerprinting is out; v2 uses behavioral BLE/AP fingerprints and AP clock skew. (4) Voice anti-spoofing is weak in the wild (ASVspoof 5 best closed EER <15%, baselines >29%); voice is a low-weight stack member, never a verdict. (5) Offline operation and remote wipe conflict; the documented default is crypto-erase triggered locally or by a paired phone, with no cloud relay.
Cost (estimates, before tax/ shipping)	Tier A v2 ≈ **$461** (v1: $344; +$117 for eye camera, mic, GPS, haptics, lens, LEDs, buttons). Tier B v2 increment ≈ **$401**, total ≈ **$862** (v1: $722). The Android phone and a Mac/Linux build machine are assumed already owned.
Ethics and legal guard (unchanged, extended)	The system **never labels a real person "robot"**, "fake" or "not alive"; it reports signatures and indicators that were observed or not observed, with benign causes. Eye, voice and liveness data are treated as **sensitive biometrics**: consenting adults only, on-device only, derived traces only, 30-day retention, written release (Section 14). Anti-tamper features protect the owner's own data on the owner's own device.
How do we know it worked?	Frozen pass/fail thresholds for 29 experiments (E-01…E-29), including headless replay, a golden dataset protocol, security tests, and per-module gates. A module that misses its gate ships **disabled by default**; it never blocks the G1 decision unless listed there.
What still needs the owner?	Only: purchase approval (D-01), and five hardware or policy facts (R-1…R-5, Section 15). Everything else is settled.

Changelog v1 → v2

Area	v1	v2
Gap audit	30 rows	same 30 rows (status updated where v2 changed them) + **32 new rows** N-01…N-32
Modules	none	**15 modules** M-01…M-15 with feasibility, design, defaults, licenses, tests
Fusion	6 scores, logistic regression	**11 scores in 6 families**, stacked logistic regression with missing-family mask, ≥2-family evidence rule, signatures A1–A11
Parts / BOM	Tier A $344 / B $722	adds eye camera + lens, mic, GPS, haptics, buttons, optional secure element, speaker; **$461 / $862**; all live prices dated
Wiring / power	GPIO18 LED gate; ~8 W / 11 W	**GPIO map redone** (I2S needs GPIO18–21; GPS moved to I2C); ~9.9 W / 12.4 W typical, 15 / 19 W peak
Platform	rig first	unchanged; **Android operator app** (new) and fallback node; Kotlin + Python share JSON score schema
Data and privacy	no audio, no GPS, no biometrics	mic, GPS and eye data added with push-to-talk, hardware mute, coarse exports, derived-only storage
Tests	E-01…E-10	**E-11…E-29** added; gates G0, G1 kept; per-module gates M-gates added
Decisions	7 defaults, 8 open	**settled defaults D-02…D-45**; residual R-1…R-5
CI / build	none	local `build_all.sh`; **no paid Actions** (macOS runners $0.062/min on private repos); macOS job is spec-only

1. Gap audit: before and after (all 30 v1 rows kept, 32 new rows)

Legend: "Closed" = researched and cited; "Decision" = closed by a recorded default; "Open" = needs hardware testing. v1 rows are unchanged except for a bold **v2:** note where v2 changed their status. **Section numbers inside v1 rows refer to the v1 PDF;** v2 equivalents: BOM 6, wiring/safety 7, software 8, data 9, build 10, tests 11, false positives 12, risks 13, legal 14, decisions 15.

A. v1 rows (before, with v2 updates)

ID	Gap, weak claim or missing item in v1	Sev.	How it is closed in this spec	Status
G-01	**Brilliant Frame listed as an option; it is delisted.** Product page 404s; no end-of-life notice (Talaya, Extentos, oculos.io).	High	Removed from options. Halo is the only Brilliant product ("Shipping now" per store).	Closed
G-02	**Halo maturity and US availability unclear.** Store says "Shipping now" but gives no date. Maker posts (reported on Reddit) show batch shipping, production pause until Oct 12, early pre-orders still waiting; return costs $49 restocking + shipping.	High	Halo moved to Phase 2 only. No Phase 0–1 dependency on it. Order-risk noted in Section 4.	Closed
G-03	**Halo "open" claim too broad.** The firmware repo is public, but the Alif SDK portion is under a vendor license (restricts use to Alif silicon; no copyleft relicensing); Brilliant docs still say repo "coming soon" while GitHub shows it.	High	Treated as "mostly open, not fully OSI-open." Decision D-06 asks the owner whether that is acceptable. **v2:** D-06 settled: Halo stays out of the default build.	Decision
G-04	**Halo camera/NPU capability uncertain.** v1 implied on-device ML. Docs show a Lua API returning 640-px JPEGs over BLE (16–80 KB each) and a raw-read call; a July review said Lua exposes no NPU/inference API.	High	Spec assumes **phone/laptop-side** processing, about 1 frame per second or less over BLE **[UNVERIFIED rate]**. On-device ML is not planned.	Closed (assumption)
G-05	**Mentra Live ordering and limits unverified.**	Med	Verified: US-address web orders only; $449; Apache-2.0 OS, SDK and Bluetooth starter kit; camera photo upload to own webhook; streaming to own RTMP/SRT/ WebRTC endpoint. Stock status **[UNVERIFIED]**.	Closed
G-06	**Meta glasses treated as an option without license check.**	Med	Verified not open source: limited, revocable license; one third-party app at a time in developer mode. Excluded by the open-source-only rule.	Closed
G-07	**Thermal parts mis-ranked.** v1 called the MLX90640 "cruder" without noting accuracy.	Med	Datasheet: ±1 °C typical vs Lepton 3.5 typical ±5 °C (±10 °C low gain). Tier A uses MLX90640 for absolute temperature; Tier B adds Lepton/UVC for spatial detail. Both tested in E-03.	Closed
G-08	**No exact SKUs, links or live prices.**	High	Section 5 BOM with links and dated prices; stock caveats (Adafruit MLX90640 out of stock; RTL-SDR V4 end-of-line). **v2:** BOM re-priced and extended (Section 6).	Closed
G-09	**IR LED eye-safety not addressed.**	High	IEC 62471 design rules and calculation (Section 6.3); exempt-group parts only; hardware current limit; no optics; consenting subjects. Independent lab verification **not** done.	Closed + caveat
G-10	**Battery safety near the head.**	Med	Battery on belt, certified pack with protection; inspection rules (Section 6.4).	Closed
G-11	**Interfaces and power budget missing.**	High	Pinout, buses, cable lengths, budget and runtime estimate (Section 6). Needs measurement (B-10). **v2:** GPIO18 conflict with I2S found and fixed; new budget (N-28, N-29).	Closed (estimate)
G-12	**No software stack, license policy or repo layout.**	High	Section 7: open-source allow-list, deny-list (AGPL, non-commercial), automated license audit, repo tree.	Closed
G-13	**No data schema or privacy design.**	High	Section 8: SQLite schema, hashing of radio identifiers, no face storage by default, 30-day retention. **v2:** schema extended for audio, GPS, eye data (Section 9).	Closed
G-14	**Success metrics not defined before testing.**	High	Section 10: frozen thresholds, gates G0/G1, rule against moving goalposts.	Closed
G-15	**No controls plan.**	High	Mannequin (cold/heated), powered toy robot, humans, phones, silicone prop, hot objects, silent devices (Section 10).	Closed
G-16	**False-positive handling vague.**	High	Operating point set at ≤5% false-positive rate on human controls; contributing signatures shown; sensor-health suppression (Section 11).	Closed
G-17	**Ethics rule only stated, not enforced.**	High	Vocabulary guard in code + automated test; output schema has no "robot" class field (Sections 2, 8, 11). **v2:** guard now also covers voice phrases and signatures A7–A11.	Closed

ID	Gap, weak claim or missing item in v1	Sev.	How it is closed in this spec	Status
G-18	**Legal section was generic; FCC claim came from a search summary.**	High	Checked FCC jammer page (jamming illegal; no personal exemptions), 18 U.S.C. §2511(2)(g), 47 CFR §15.121 via search results; checklist in Section 13. The FCC "interception" consumer guide fetch timed out, so wording is **[UNVERIFIED at source]**. **v2:** legal analysis extended to biometrics, voice, GPS (Section 14).	Closed + caveat
G-19	**Thermal claims could drift into health claims.**	Med	FDA's thermography guidance expects ≤±0.5 °C and a blackbody reference for fever triage; our rig is not a medical device and never reports body temperature as health data.	Closed
G-20	**"Phones carry cameras" false-positive for lens glint** and human eyes also retroreflect.	Med	Test E-01 includes phones, glasses, eyes; glint score uses IR-on minus IR-off differencing.	Closed (to be measured)
G-21	**Wi-Fi scanning scope ambiguous.** Monitor mode and probe-request capture raise legal and privacy questions.	Med	Phase 0–1 = BLE advertisements and AP beacons in managed mode only; no monitor mode, no data frames, no cellular. Decision D-05. **v2:** AP clock skew read from managed-mode scan TSF; still no monitor mode.	Decision
G-22	**mmWave module integration unproven.** MR60BHA2 ships with ESPHome for Home Assistant; direct Pi use not verified.	Med	Treated as integration risk RK-07; fallback: read via ESPHome native API or USB serial; drop to Tier B-lite if blocked.	Open
G-23	**Memory-EM (ESauron-style) idea untested with cheap SDR.** Paper used USRP B210.	Low	Optional exploratory experiment E-10, not a gate. **v2:** RTL-SDR V4 range now verified (500 kHz–1.766 GHz); E-10 stays exploratory.	Open
G-24	**Mounting, weight, enclosure absent.**	Med	Pod target <120 g, compute on belt, cable lengths, strain relief (Section 6.2).	Closed
G-25	**No calibration or timing design.**	Med	Thermal two-point calibration, NoIR/thermal alignment, clock sync, drift limits (Section 9).	Closed
G-26	**No logging/reproducibility.**	Med	Session manifest, config hash, git commit, sensor serials in every run (Section 8).	Closed
G-27	**Human-subject policy absent.**	High	Adult household volunteers, written consent form, right to delete, no strangers (Sections 10, 13). Decision D-04. **v2:** D-04 settled (the owner first, then household adults).	Decision
G-28	**other projects coupling.**	Low	Separate repo and data; no shared code or models.	Closed
G-29	**Robot-specific detection performance has no published evidence.**	High	Stated as hypothesis; Phase 1 can only show separability of lab proxies, not real humanoids. Real-robot test is Decision D-08.	Open
G-30	**Cost bands were guesses.**	Med	Two tiers ($344 / $722 estimated, excluding tax/shipping); alternatives listed. **v2:** re-priced: $461 / $862.	Closed

B. New rows found in the v2 re-audit and by the 15 modules

ID	New gap found in the v2 re-audit or by a module	Sev.	How it is closed in this spec	Status
N-01	**No eye-behavior physics in v1.** Micro-saccade, pupil and corneal signals were never assessed for distance, frame rate or light needs.	High	Section 5 M-01: ≥250 Hz needed for micro-saccades (sources) vs ≤120–143 fps cameras; pupil shift ≈0.09 mm ≈0.5 px at 0.5 m (my calculation). Micro-saccades: report-only. Pupil size, blink, glint geometry at 0.3–1 m: feasible. E-11…E-14.	Closed (analysis); tests open
N-02	**v1 forbade face detection, but eye analysis needs an eye region.**	High	Operator-aimed narrow eye camera; transient eye-region detection in RAM only; no face templates, no stored face crops (D-17).	Decision
N-03	**v1 non-goal 'no biometric' no longer true:** eye dynamics, voiceprint and liveness are biometric-adjacent.	High	Legal analysis (Section 14): BIPA lists retina/iris scans and voiceprints; WA RCW 19.375 and TX CUBI cover voiceprints; GDPR Art. 9. On-device only, derived traces, written release, 30-day retention, deletion rebuilds the golden set.	Closed + caveat
N-04	**v1 safe harbor 'no microphones' lost** once voice is added; all-party-consent states.	High	Push-to-talk lab prompts only, hardware mute switch, no ambient recording, enrolled consenting speaker only (D-18). CA, FL, IL, WA are all-party states (recordinglaw.com).	Decision
N-05	**Pupillary light reflex needs controlled light** and smartphone pupillometry fails often (30% vs 4% for infrared devices).	Med	Passive pupil/hippus by default; active light step only in lab, with photometer and consent (D-15). E-13 is report-grade.	Decision
N-06	**Silicone mask on a live wearer keeps real eyes, pulse and voice.**	High	Stated limit: eye, pulse and voice signals cannot catch it; only thermal/NIR skin cues might (E-15 reports it, no pass criterion).	Closed (limit)
N-07	**Liveness signals are spoofable:** rPPG, blink, voiceprint can be replayed or cloned.	High	Stack of independent families, challenge-response IR glint code, no standalone verdict, ROC reported with attack set.	Closed (design); tests open

ID	New gap found in the v2 re-audit or by a module	Sev.	How it is closed in this spec	Status
N-08	**Voiceprint purpose creep:** liveness consistency vs identifying people.	High	Purpose limitation: lab liveness checks on enrolled volunteers only; never used to identify strangers; embeddings ephemeral unless written enrollment (D-18).	Decision
N-09	**v1 implied SDR fingerprinting; RTL-SDR V4 tunes only 500 kHz–1.766 GHz.**	High	IQ-level BLE/Wi-Fi RFFI out of Phase 0–1. Behavioral BLE fingerprints + AP clock skew from beacon TSF (Jana and Kasera, MobiCom 2008) instead. A 2.4 GHz SDR is a residual item (R-3).	Closed (correction)
N-10	**BLE fingerprint limits:** random addresses rotate; clocks allow ±50 ppm active / ±500 ppm sleep; IQ fingerprints degrade at low SNR and in motion.	Med	Baselines only for fixed emitters on own premises; 'unknown' is a valid class; no claim of tracking phones.	Closed
N-11	**'Tampering' is an over-claim:** drift can be firmware, temperature, battery.	Med	Output 'emitter changed vs baseline' (A9), needs ≥2 changed features persisting ≥3 windows; no attribution of cause.	Decision
N-12	**Security was LUKS only:** no key hierarchy, channel, or log integrity.	High	M-04: AES-256-GCM/XTS at rest, Keystore keys, Noise_XX BLE channel, hash-chained HMAC log, anchors.	Closed (design)
N-13	**Offline operation conflicts with remote wipe.**	High	Resolved default D-23: crypto-erase triggered locally (panic, failed unlocks) or by a paired phone in BLE range; optional user-owned relay OFF by default; dead-man timer OFF.	Decision
N-14	**Overwrite wipe is unreliable on SD/flash (wear levelling).**	Med	Wipe = destroy keys (LUKS header erase, Keystore entry delete), not overwrite.	Closed
N-15	**GPS privacy and v1 'No GPS' field.**	Med	Encrypted at rest, rounded to 0.01° on export, indoor 'no fix' recorded not guessed (D-25).	Decision
N-16	**OSM tile policy:** bulk or pre-seeded downloads from tile.openstreetmap.org are forbidden; ODbL attribution needed.	Med	Offline maps from Mapsforge map files built from OSM extracts; attribution on screen; no tile-server fetches (D-26).	Closed
N-17	**TTS licensing:** Piper (piper1-gpl) and eSpeak NG are GPL-3.	Med	Excluded from the default build. Platform TextToSpeech with offline voice on the phone; Flite (BSD) on the Pi; sherpa-onnx (Apache-2.0) optional once a voice's license is read (D-27).	Closed
N-18	**'Threat-tiered' wording conflicts with the ethics rule.**	Med	Tiers are named FYI / Attention / Confirmed and refer to *observed signatures*, not people; Confirmed requires operator confirmation; phrases come from a whitelist (D-28).	Decision
N-19	**Per-location baselines are hard for a moving head-mount.**	Med	RF-first baseline; thermal/NIR comparison only at marked anchors (ArUco, OpenCV); report-only thresholds for the optical path (M-07, E-21).	Closed (limit)
N-20	**Color thermal palettes mislead and imply medical readings.**	Low	Grayscale default, fixed scale with legend, overlay on request (M-08, E-22).	Closed
N-21	**No headless mode or regression data:** tests needed hardware and people.	High	Replay/sim adapters, virtual clock, golden dataset with hash manifest and frozen test split (M-09, E-23).	Closed (design)
N-22	**macOS CI would cost money:** standard macOS runners $0.062/min on private repos (GitHub docs, 2026-10-28).	High	No paid Actions. Local `scripts/build_all.sh`; optional free self-hosted runner; CI job is spec-only (D-32).	Decision
N-23	**macOS signing and notarization need a paid Apple program** [UNVERIFIED fee].	Low	macOS artifact is an unsigned local build; documented Gatekeeper step; no notarization.	Decision
N-24	**Sensors were wired into feature code:** no abstraction for replay, simulation or the phone.	High	M-11 adapter contract; real, replay and sim adapters; Kotlin adapter for the phone.	Closed (design)
N-25	**Feedback loop could contaminate the pre-registered test set.**	High	Quarantine → review → growth set; frozen test split is immutable and hash-checked (M-12, E-26).	Closed
N-26	**Android fallback could depend on proprietary Google services** (ML Kit, Play Services location).	Med	AOSP APIs only (LocationManager, CameraX, BiometricPrompt); no ML Kit, no Firebase; minSdk 29 (M-13).	Closed
N-27	**App gate can be a UI-only check that is bypassable.**	High	Biometric unlock bound to a Keystore key (CryptoObject); BIOMETRIC_STRONG only; FLAG_SECURE (M-15, E-29).	Closed
N-28	**Pin conflict:** v1 LED gate on GPIO18 (an I2S clock pin); v2 needs I2S, GPS, haptics.	High	GPIO map redone (Figure 3); GPS on I2C instead of UART.	Closed
N-29	**Power budget omitted new loads.**	Med	New budget: ~9.9 W / 12.4 W typical, peak 15 / 19 W; 25 W PD profile required.	Closed (estimate)
N-30	**License status of new libraries unchecked.**	High	Verified: Tink Apache-2.0; SQLCipher BSD-3; noise-java and Python noiseprotocol MIT; AASIST MIT; ECAPA model Apache-2.0; sherpa-onnx Apache-2.0; MapLibre Native BSD-2; Mapsforge LGPL-3 with shipping exception. Excluded: Piper, eSpeak NG, CryptoAuthLib (non-OSI). Model *training-data* licenses are **[UNVERIFIED]**.	Closed + caveat

ID	New gap found in the v2 re-audit or by a module	Sev.	How it is closed in this spec	Status
N-31	**Tiny samples:** ~150 runs cannot support many fusion inputs.	Med	≤6 stacking inputs (one per family), L2 penalty, bootstrap CIs clustered by subject, report-only for sub-groups.	Closed
N-32	**Anti-tamper and wipe features can be misread as evidence destruction.**	Med	Scope: protects the owner's own data on the owner's own device; nothing in Phase 0–1 involves third-party records; preservation duty noted (Section 14).	Decision

2. Scope, requirements and non-goals

Purpose. Learn, with honest measurements, how much separable signal exists between living people and powered or heated stand-ins using low-cost passive sensing; and provide a **hidden-camera sweep** tool for the owner's own premises, with an encrypted sighting log. Phase 0–1 only: private premises, consenting adults, no strangers, no public places.

Requirements. R-01…R-10 from v1 stand (R-03 now covers two LED banks; R-07 now covers audio, GPS and eye traces). New:

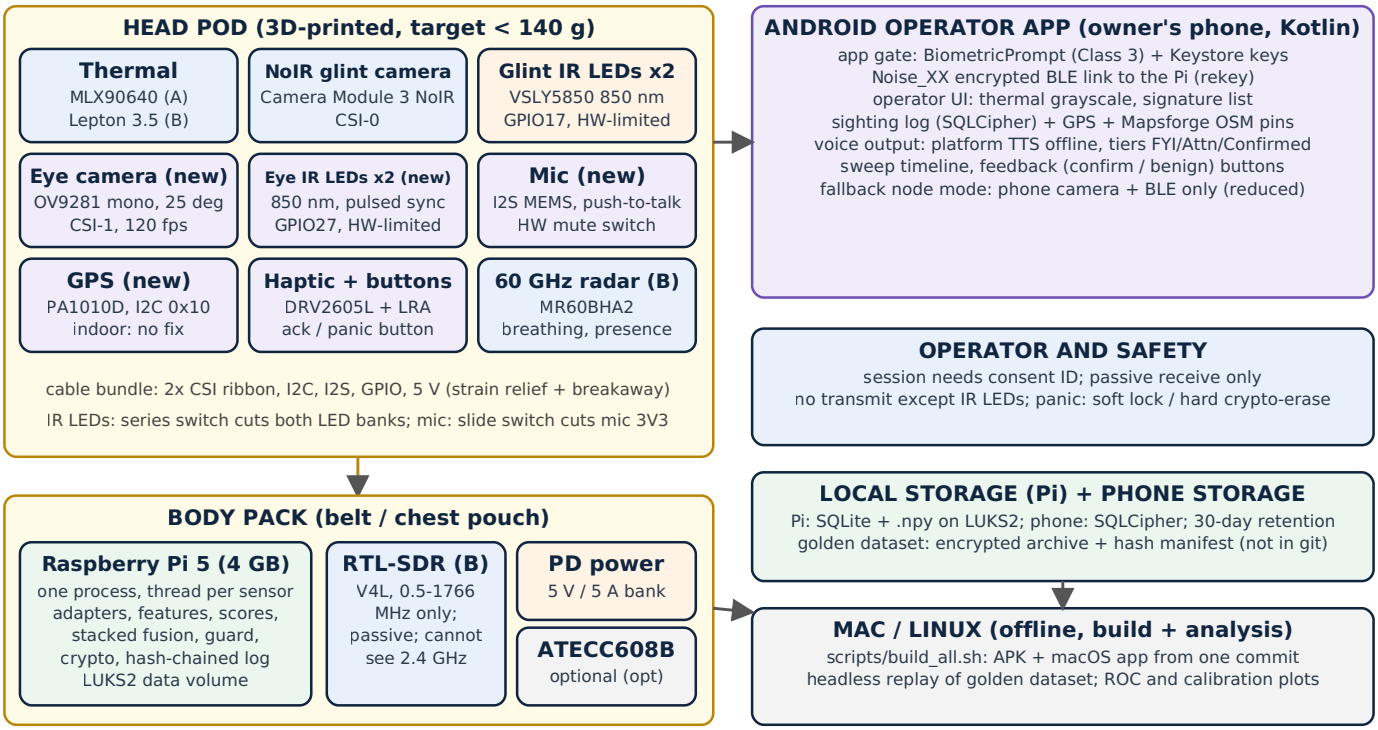
ID	Requirement	Verified by
R-11	Eye module outputs pupil, blink and glint-geometry features at ≤1 m; micro-saccade output is offline report-only.	E-11…E-14
R-12	Liveness stack needs ≥2 independent families to raise an indicator-absent signature; never a verdict.	E-15, E-16
R-13	Fingerprint module baselines fixed emitters, flags drift (A9) after ≥2 features change for ≥3 windows.	E-17
R-14	Data at rest AES-256 (GCM or XTS); keys never in plaintext on disk; constant-time comparisons; channel encrypted and rekeyed; log hash-chained.	E-18
R-15	Works fully offline; remote wipe resolved by D-23; panic gesture locks in ≤1 s.	E-18
R-16	Sighting log is encrypted, auto-stamps GPS when a fix exists, shows pins on an offline map with OSM attribution, no network access in the offline build.	E-19
R-17	Voice output has three tiers; Confirmed barges in within ≤300 ms; only whitelisted phrases.	E-20
R-18	Per-site baselines; new fixed emitter or object at a known anchor raises A10.	E-21
R-19	Thermal display is grayscale by default; color overlay is a toggle.	E-22
R-20	Whole pipeline runs headless from recorded data with a virtual clock and a frozen golden dataset; identical outputs on repeat.	E-23, E-25
R-21	One commit builds an Android APK and a macOS artifact via a local script; no paid CI.	E-24
R-22	Confirmed sightings become quarantined candidate clips; the frozen test split never changes.	E-26
R-23	Any Android device with a camera can run as a reduced detection node.	E-27
R-24	Sweep timeline shows what is new since the previous sweep; records are tamper-evident.	E-28
R-25	App opens only after strong biometric or device-credential unlock; keys are bound to that unlock.	E-29

Non-goals. No face recognition or face templates; no identification of any person; no stored face crops; no strangers; no public spaces; no cloud services; no cellular, no monitor mode, no jamming, no transmit except IR LEDs and the Pi/phone's normal client radios; no medical readings; no claim that output identifies a robot; no use of the security features to hide wrongdoing or evade lawful requests.

Standing rules. Open-source only (no AGPL, no non-commercial licenses), clean temp files, never touch other projects, nothing purchased or sent without the owner's approval, **no paid GitHub Actions**.

3. System architecture and fusion design

Figure 1. System architecture v2 (pod, belt pack, Android app; B = Tier B, opt = optional)



Process model (unchanged in spirit). The Pi runs **one process** with a thread per sensor adapter, a sync layer, a 1 Hz feature/score loop, one fusion layer, and a guard. The phone app has the same shape in Kotlin (coroutines per adapter). Why a pod and belt pack: heat, weight and battery risk stay off the head; the pod holds only the sensors that need the wearer's viewpoint.

Signatures reported (vocabulary). Observations, never identities. A1–A6 are as in v1; A7–A11 are new.

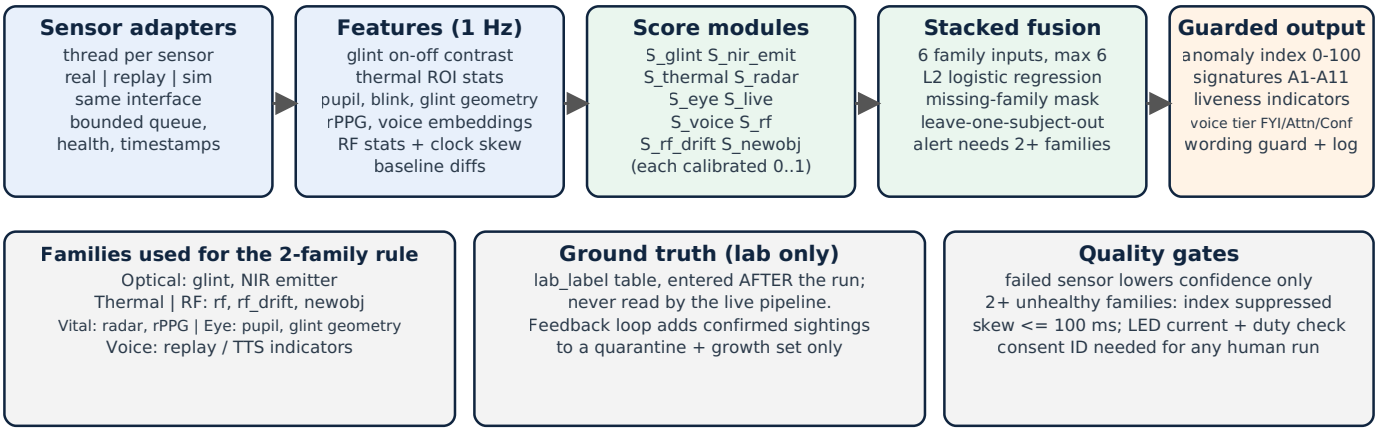
Code	Signature	Source	Typical benign causes (always shown with the alert)
A1	Lens-glint candidate	NoIR + IR LED code	Phones, webcams, glasses, watches, jewelry, human eyes
A2	Active near-IR emitter	NoIR (LED off)	Security cameras, remotes, face-unlock, night-vision rings
A3	Thermal pattern atypical vs baseline	Thermal	Cold hands, coats, hot drinks, fever, heaters, sun-warmed objects
A4	Radio emitter near	BLE/Wi-Fi	Phones, watches, earbuds, medical devices, beacons
A5	Expected vital motion absent at ≤1 m (B)	mmWave	Subject moving, wet clothing, radar blocked
A6	Motion pattern irregular	Frames + IMU	Ordinary movement, children
A7	Eye-dynamics atypical (pupil, blink, glint geometry)	Eye camera	Sunglasses, contact lenses, drops, fatigue, medication, glare, glass eye (prosthesis)
A8	Voice-liveness indicators not observed	Mic	Poor mic placement, noise, a speaker phone, a cold, a recording device in use
A9	Known emitter changed vs its baseline	BLE/AP	Firmware update, battery, temperature, relocated device
A10	New fixed emitter or object at a known anchor	Baseline diff	Guests' devices, moved furniture, new appliance
A11	Expected liveness indicators absent across ≥2 families	Stack	Photo or video on a screen, mask on a stand, mannequin, wearer outside the sensor's range

Fusion v2 (stacked, deliberately small).

1. Each module emits a score in [0, 1], a confidence in [0, 1] and a health flag (unchanged contract). Eleven scores: S_glint, S_nir_emit, S_thermal, S_rf, S_radar, S_motion, S_eye, S_live (rPPG + blink), S_voice, S_rf_drift, S_newobj; adapter health is a flag, not a score.
2. Scores fold into **six families** (optical, thermal, RF, vital, eye/liveness, voice): the family value is the confidence-weighted maximum of its member scores. Level 2 is one L2-regularized logistic regression (scikit-learn, BSD-3) over ≤6 family values with a **missing-family mask**; scores are cross-fitted so the level-2 model never sees scores it trained on.
3. Evaluation is leave-one-subject-out, with bootstrap confidence intervals clustered by subject (N-31). Threshold is set at human false-positive rate ≤5%; the true-positive rate is reported even if low.
4. Evidence rule: an alert needs ≥2 independent families; one family alone is an "FYI". If ≥2 families are unhealthy the index is suppressed.

5. **Outputs:** `anomaly_index`, `signatures[]`, and `liveness_indicators[]` (each "observed" or "not observed", with the benign-cause list). No output field names a class.

Figure 2. Adapters, score modules and stacked fusion (scores 0..1; output never labels a person)



4. Platform decision: rig first, glasses later (facts verified 2026-10-28; unchanged from v1)

Platform	Verified facts	Fit
Head-mounted rig (Pi 5 + sensors) + Android operator app	Only route to thermal, IR LED, radio and eye sensing; every part has a datasheet or store page.	**Chosen**
Mentra Live + MentraOS	$449; US-address web orders only; 43 g, 12 MP camera; MentraOS, Bluetooth SDK and starter kit Apache-2.0; no display. IR sensitivity, raw IMU **[UNVERIFIED]**.	Phase 2 carrier; spike S-01 optional
Brilliant Labs Halo	$399, batch shipping; Lua API returns 640-px JPEGs over BLE; firmware public with mixed licenses (Alif SDK vendor license).	Phase 2 only; D-06 settled: not in the default build
Brilliant Labs Frame	No longer sold; no end-of-life notice.	Excluded
Meta Ray-Ban/Oakley	Not open source; limited revocable license; one third-party app at a time.	Excluded
Even G2, Vuzix Z100, Snap SPECS, XREAL, Android XR	No camera, closed, or not shipping.	Excluded / watch

Glasses compatibility spike S-01 (does a glasses camera even pass 850 nm?) stays optional and skipped until G1 (D-07 settled: skip).

5. Module specifications M-01…M-15

Each module follows one template: **Feasibility** (what the physics and literature allow), **Design** (what we build, with settled defaults), **Libs** (license checked), **Gate** (the experiment that decides whether it stays enabled). Calculations marked "(calc)" are mine, from the stated assumptions, not measurements.

M-01 Eye-behavior analysis (signals S_eye; signature A7)

Feasibility. Micro-saccades last a few ms and studies use ≥250 Hz trackers; 30–60 Hz webcams miss them, 120 Hz still under-samples (Tobii/PMC, RemoteEye). Our best eye camera (OV9281, 120 fps USB / up to 143 fps CSI per the vendor listing) is below 250 Hz. A 0.5° eye rotation moves the pupil ≈0.09 mm (calc, assuming a ≈10 mm lever arm). With a 25° lens on 1280 px the pixel footprint is 0.17 mm at 0.5 m and 0.35 mm at 1 m (calc), so the shift is ≈0.5 px and ≈0.25 px. **Micro-saccade detection at a distance is not feasible; it is report-only.** Pupil diameter is feasible: a 4 mm pupil spans ≈23 px at 0.5 m, ≈12 px at 1 m, ≈6 px at 2 m (calc). The pupillary light reflex (latency ≈0.17–0.23 s) needs controlled light, and smartphone pupillometers failed 30% of attempts vs 4% for infrared devices. Hippus (0.3–0.7 Hz) is too small to rely on. Corneal reflections work as liveness cues against photos, screens and some masks (Y-EYES/AIME report ≈200 ms detection), but their reliability against contact lenses and hand-made prosthetic eyes is unclear and ocularist-made eyes can look live even in near-IR; only central specular reflections hint at an odd cornea. **A wearer under a silicone mask keeps real eyes.**

Design. (1) Operator-aimed OV9281 with a 25° M12 lens at 0.3–1 m; two 850 nm LEDs pulsed in sync with exposure; 400×300 eye ROI found transiently in RAM (OpenCV Haar eye cascade, Apache-2.0 [cascade-file license UNVERIFIED]); no face template, no stored crops (D-17). (2) Features at 60–120 fps: blink rate and duration, pupil diameter trace (ellipse fit), glint count/size/offset from pupil centre, bright-pupil response (on-axis LED) and a **glint challenge-response**: the two LEDs run a random on/off code; a live cornea's glints must follow the code within one frame, a replayed screen cannot. (3) Camera-lens glints (A1) look different from corneal ones: sharp, tiny, no pupil structure, retroreflective; the eye module can down-weight them. (4) Active light-step PLR: lab only, consenting volunteer, visible LED ≤1 s with a photometer reading logged, start dim; default OFF (D-15). (5) Micro-saccade estimator runs offline on saved 120 fps pupil traces for the report only. **Libs.** OpenCV (Apache-2.0), NumPy/SciPy (BSD-3). **Gate.** E-11…E-14; if E-11 or E-14 fail, S_eye ships disabled.

M-02 Liveness detection with multi-signal stacking (S_live, S_voice; signature A11)

Feasibility. rPPG pulse works at short range with still subjects and is spoofable by lighting and motion; mmWave breathing helps to 1.5 m (v1 evidence). Voiceprint (speaker embedding) can verify an *enrolled* volunteer but a cloned voice passes it; liveness needs a separate anti-spoof model, and ASVspoof 5 shows top closed-set EER <15% while baselines exceed 29%: weak in the wild. A live TTS speaker cannot be told from a person by voice alone. Therefore no single family decides anything. **Design.** Families: Optical/thermal (skin-region thermal texture + NIR response), Vital (rPPG from NoIR/eye cam, mmWave breathing), Eye (blink, pupil, glint code), Voice (replay/TTS indicators), RF (a person's carried devices; absent devices prove nothing). Stack per Section 3: ≥2 families must agree that expected indicators are *not observed* to emit A11, wording "Expected liveness indicators not observed: pulse, blink. Benign causes: ...". Challenge-response parts: glint code (eye), a spoken digit prompt (voice, lab only). **Libs.** Own implementation of published rPPG methods (CHROM/POS patent status **[UNVERIFIED]**); scikit-learn. **Gate.** E-15.

M-03 Signal fingerprint module (S_rf, S_rf_drift; signature A9)

Feasibility. Physical-layer BLE fingerprints exceed 99% in lab studies but degrade with low SNR, motion, power level and temperature (Oakland 2022: viable but sometimes unreliable in a field study of hundreds of devices; WCNC 2025: a noise-robust model still keeps only ≈75% at 10 dB SNR), and they need raw IQ, which the Pi's BLE chip and an RTL-SDR (500 kHz–1.766 GHz) cannot provide. BLE clocks allow ±50 ppm active and ±500 ppm sleep, so timing is not a unique ID. AP clock skew from beacon TSF timestamps is a proven fingerprint (Jana and Kasera, MobiCom 2008: stable per AP, estimable from ≈50–100 frames) and is available in managed-mode scans. **Design.** For **fixed emitters on the owner's own premises** only: baseline = advertising-interval mean/jitter, TX-power field, payload template and length, service UUID set, company ID, per-channel (37/38/39) ratio where a sniffer is present, RSSI distribution at anchor, and AP TSF skew. Drift (A9) = ≥2 features outside 3σ of baseline for ≥3 consecutive 60 s windows. Device-type identification = lookup on company ID, OUI of APs (not of rotating BLE addresses), service UUIDs, and payload pattern into ≥5 coarse types plus "unknown". No attempt to track phones or people; random-address devices are counted, not followed. **Libs.** bleak (MIT), `iw` as a subprocess (ISC/GPL tools only as executables), NumPy. **Gate.** E-17.

M-04 Security module (cross-cutting)

Scope: **protects the owner's own data on the owner's own device** (D-45).

Control	Default design	Library (license)
Data at rest	Pi: LUKS2 (AES-256-XTS) data volume; phone: SQLCipher DB, 256-bit key wrapped by an Android Keystore AES-256-GCM key (StrongBox when present, attestation checked, `setUserAuthenticationRequired`).	cryptsetup (GPL tool, subprocess), SQLCipher (BSD-3), Android Keystore (AOSP), Tink (Apache-2.0)
Constant-time crypto	No custom primitives. Compare secrets only with `hmac.compare_digest` / libsodium `sodium_memcmp` / Tink. A lint test bans `==` on key, tag or MAC variables. Python cannot guarantee timing of high-level code; stated as a limit.	libsodium via PyNaCl (ISC / Apache-2.0), BoringSSL inside Tink
Pod ↔ phone channel	Pair once by QR code (static key fingerprints out of band). Per session: **Noise_XX_25519_ChaChaPoly_BLAKE2s** handshake over BLE GATT; payloads are scores, events, thermal frames (≈6 KB/s); **Rekey every 1,000 messages or 10 min**, whichever first; re-handshake every session; static keys rotated every 90 days or on re-pair. A double ratchet is not used: it targets asynchronous messaging and adds complexity (D-22).	noise-java (MIT) on Android; Python `noiseprotocol` (MIT) on the Pi
Offline operation	No cloud, no INTERNET permission in the offline APK build; time from RTC/GPS.	—
Tamper-evident log	Append-only; `h_i = HMAC-SHA256(k_log, h_{i-1} ‖ canonical(record_i))`; the head hash and count are signed (Ed25519) and anchored on the phone every 60 s or 100 records; verification detects edits, deletions, reordering, and truncation past the last anchor.	hashlib (PSF), PyNaCl
Kill switch / remote wipe	**Conflict resolved (D-23):** an offline device cannot receive a remote command, so wipe is **crypto-erase** (LUKS header erase + Keystore key delete; overwriting flash is unreliable). Triggers: panic gesture, 10 failed unlocks, or a signed wipe command from the paired phone over the Noise link. An optional user-owned HTTPS relay is OFF by default; dead-man timer OFF. The phone's own loss case is covered by the OS vendor's find-my-device service, which is not part of ASB.	—
Panic gesture	Pod button: hold 1 s = **soft panic** (clear screens, drop session keys, require unlock); then press 3× within 5 s with haptic confirmation = **hard panic** (crypto-erase). Phone: 5× volume-down in the app, soft panic only. Accidental hard erase is guarded by the double step.	—

Gate. E-18.

M-05 Sighting log (S: events; map screen)

Design. Each sighting = signature codes, scores, thermal/NIR crop (non-person only), operator verdict (confirmed / benign / unresolved), GPS fix, timestamps, hash-chain link. Stored in SQLCipher on the phone; the Pi keeps a mirror on the LUKS volume. GPS: Android `LocationManager` GPS provider (AOSP, no Play Services); pod GPS (PA1010D) as backup. Indoors there is usually no fix: recorded as "no fix", never guessed. Exports round coordinates to 0.01° (≈1.1 km) unless the owner overrides (D-25). **Offline map:** Mapsforge (LGPL-3.0 with its shipping exception) renders Mapsforge map files built from OSM extracts (ODbL data); "© OpenStreetMap contributors" stays visible; the OSM Foundation tile policy forbids bulk or pre-seeded downloads from tile.openstreetmap.org, so we never fetch those; MBTiles + MapLibre Native (BSD-2) is the alternative. Sighting pins live in our own DB layer, not merged into OSM-derived data. **Gate.** E-19.

M-06 Voice output with tiered barge-in

Design. Tiers describe *observed signatures*, not people. **FYI**: haptic tick, no speech. **Attention**: ≤8-word whitelisted phrase, queued, rate-limited to one per 10 s. **Confirmed**: spoken immediately, **barges in** (cuts current speech) and repeats until acknowledged;

requires an operator confirm button, or ≥3 families plus operator ack; for sweep mode it means "operator physically found it". Phrases are templates ("Candidate camera lens at left, one metre"); no free text; the vocabulary guard checks every template at build time. TTS: Android platform TextToSpeech with an **offline voice** (reject voices that need network); Flite (BSD) on the Pi; Piper and eSpeak NG are GPL-3 and excluded; sherpa-onnx (Apache-2.0) is allowed only with a voice whose model card license is read first. Phone earpiece/wired earphones; Bluetooth earbuds would appear as RF emitters (own devices go on a self-allowlist). **Gate.** E-20.

M-07 Per-location baselines and new-object alert (S_newobj; A10)

Design. A site is a named place with 1–6 **anchors** (ArUco markers, OpenCV, Apache-2.0). Baseline pass ×3: RF set (fixed emitters), NoIR glint candidate list, thermal background (median 32×24 frames) at each anchor. Later passes compare: **new fixed emitter** (most reliable), new thermal blob ≥5 °C above ambient and ≥4 px, new glint candidate. **Limits (honest):** a hand-held head view does not register exactly; lighting, moved furniture and warm sunlit surfaces cause misses and false alarms; optical A10 thresholds are report-only, RF A10 is gated. **Gate.** E-21.

M-08 Thermal grayscale default with color overlay toggle

Default view: linear 8-bit grayscale on a fixed 10–45 °C scale (legend always shown), so ranking is by brightness and no palette implies a threshold; toggle = inferno/viridis-style overlay (matplotlib palettes are permissive-licensed) alpha-blended on the NoIR frame; auto-range toggle separate. The toggle never changes stored data (raw arrays are always saved) and a "not a medical measurement" tag remains. **Gate.** E-22.

M-09 Headless test mode and golden dataset

Design. `asb --headless --replay <clip-set>` runs the full pipeline with **replay adapters** (recorded timestamps), a **virtual clock**, GPIO/LED stubs that assert the LED is never driven, no network and no hardware. Golden dataset = versioned clips (derived features and non-person NoIR/thermal arrays; volunteers' raw eye/voice never included) with `manifest.json` (SHA-256 per file, label, source, consent ID, license, split). Splits are subject-disjoint: **train**, **dev**, and a **frozen test** split whose manifest hash is recorded in the decision log at G0; the frozen split never changes (re-freeze needs a new decision and resets claims). Stored as an encrypted archive on local disk; git holds only manifests and code. Runs on Linux (box or laptop), Pi and Mac. **Gate.** E-23, E-25.

M-10 macOS CI job producing the Android APK and macOS artifacts from one commit (spec-only)

Decision (D-32). GitHub-hosted macOS runners are free on public repos but cost **$0.062 per minute on private repos** (GitHub docs, 2026-10-28); the owner's rule forbids paid Actions. Therefore the "CI job" is a **local script**: `scripts/build_all.sh` (run on the owner's Mac) refuses a dirty git tree, records `git rev-parse HEAD`, builds the APK (Gradle, JDK 17, Android command-line tools; the Android SDK license is Google's and is a build-time tool only) and the macOS app (py2app, MIT, or a zipped CLI), writes `BUILD_INFO.json` (commit, tool versions, SHA-256 of both artifacts) into both, and verifies the two commit hashes match. Linux alternative: the same Gradle APK build on the box (free). Railway Linux builders are possible but not priced here **[UNVERIFIED]**. An optional free self-hosted runner (GitHub states self-hosted usage is currently free; a planned $0.002/min fee was postponed, so recheck) may call the same script; hosted macOS only if the repo is public and the owner approves. macOS artifact is **unsigned**; notarization needs a paid Apple program (fee **[UNVERIFIED]**). **Gate.** E-24.

M-11 Sensor bus / driver adapter layer

Contract (Python `Protocol`; Kotlin `interface` mirrors it from one JSON schema).

```
class SensorAdapter(Protocol):
    id: str; bus: Literal["i2c","csi","usb","uart","i2s","gpio","ble","wifi","android"]
    caps: set[str]                    # e.g. {"thermal_frame","nir_frame","rf_adv","audio_ptt"}
    def open(self, cfg: dict) -> None; def start(self) -> None; def stop(self) -> None
    def read(self, timeout_s: float) -> Sample | None    # Sample{t_mono_ns,t_wall,seq,units,quality,data}
    def health(self) -> Health                           # OK | DEGRADED | FAILED + reason
class ScoreModule(Protocol):
    def update(self, window: Window) -> Score            # Score{value 0..1, confidence 0..1, health, evidence}
```

Rules: one thread per adapter; bounded queue with drop-oldest and a drop counter; timestamps stamped at read time with monotonic ns; a crash sets FAILED and never stops the session; USB devices re-open with back-off (≤10 s); capability registry lets fusion mask missing families (this is how the phone fallback works). Variants per interface: **real**, **replay** (clip files), **sim** (synthetic generators for unit tests). Adapter and score plug-ins declare a license field; the audit rejects non-allowed licenses. **Gate.** E-25.

M-12 Feedback loop: confirmed sightings → labeled golden clips

Design. Operator marks a sighting *confirmed* (e.g. camera physically found) or *benign* (false positive). The app packages ±10 s of derived features and non-person arrays into a **quarantine** candidate with label, site, consent flag, and hash-chain link. A reviewer (the owner) promotes it to the **growth set** (used for train/dev only). Rules: frozen test split never receives clips; any clip with a person but no consent ID is rejected automatically; withdrawn consent deletes the clip and triggers a manifest version bump; models are retrained only offline with logged seeds. **Gate.** E-26.

M-13 Fallback mode: any Android device with a camera as a detection node

Design. The Kotlin adapters (CameraX, BLE scanner, optional mic) run with the same score modules via the capability registry. Available: A1 glint by white-flash on/off differencing (phones lack an IR illuminator and most front cameras have an IR-cut filter), A4 BLE scan, optional blink/pupil coarse features from the front camera (30–60 fps, no micro-saccades). Not available: thermal, radar, IR eye illumination. Constraints from Android: BLE scan start/stop limited to 5 per 30 s; Wi-Fi scans 4 per 2 min in the foreground, 1 per 30 min in the background; Android 12+ needs BLUETOOTH_SCAN (use `neverForLocation` only if we do not derive location from

scans, which filters some beacons). **A fallback node can never alert alone** (only 2 families: optical + RF); output is advisory. AOSP/Apache libraries only: CameraX, OpenCV Android, TFLite (Apache-2.0); no ML Kit, no Play Services, minSdk 29 (D-35). **Gate.** E-27.

M-14 Sweep history timeline

Design. A **sweep** = one hidden-camera session at a site. Table `sweep` (site, start/end, operator, duration, counts per signature, outcomes, GPS, anchor hash). Timeline screen lists sweeps and shows "new since last sweep" using M-07 baselines; records are hash-chained and exports are encrypted. **Gate.** E-28.

M-15 App gate

Design. Phone: `BiometricPrompt` with **BIOMETRIC_STRONG** (fingerprint, or face only if the device rates it Class 3) with device-credential fallback; the unlock is bound to a Keystore key through a `CryptoObject`, so the database cannot be decrypted by bypassing the UI check (OWASP warns UI-only checks are bypassable); auto-lock after 60 s in background; `FLAG_SECURE` blocks screenshots and recents thumbnails; OS-enforced lockout after repeated failures; 10 failed credential attempts trigger D-23 crypto-erase. The app never sees biometric data (the OS keeps it), so the gate itself is not a collection of biometrics by ASB (not legal advice). Mac tool: Keychain item with biometric access control. Pi: no screen gate; physical button + LUKS passphrase at boot (typed, or delivered by the paired phone). AndroidX Biometric is Apache-2.0. **Gate.** E-29.

6. Bill of materials v2 (two budget tiers)

Prices are list prices from store pages or search listings seen **2026-10-28**, USD, **before tax and shipping**. "Est." = my estimate, **[UNVERIFIED]**. Stock not confirmed unless stated. **Not included:** the Android phone (the owner's own, must support Keystore; StrongBox and Class 3 biometrics preferred), the Mac/Linux build machine, and any mannequin or props. Nothing here has been ordered.

Tier A v2: bench rig + eye, voice, GPS, haptics (v1 rows A1–A12 unchanged; A13–A20 new).

#	Item	SKU / link	Qty	Price	Notes and alternatives
A1	Raspberry Pi 5, 4 GB	raspberrypi.com/products/raspberry-pi-5/	1	$110	Price rose $25 in April 2026 (memory costs). 8 GB $175. Alt: Pi 4 4GB $100 (slower). Pi Zero 2 W ($15) is widely sold out.
A2	Official 27 W USB-C supply (bench)	raspberrypi.com/products/27w-power-supply/	1	$12 est.	Price not seen; needed for 5 V/5 A and USB current.
A3	Active cooler	Raspberry Pi	1	$5 est.	Pi 5 can reach ~12 W.
A4	Camera Module 3 **NoIR** (75°)	adafruit.com/product/5659	1	$25.00	In stock (27) at Adafruit when checked; Pimoroni NoIR "coming soon". Alt: Pi GS camera (global shutter, price not checked).
A5	Pi 5 camera cable (22→15 pin), 300 mm	Raspberry Pi / resellers	1	$5 est.	Pi 5 needs the mini CSI cable.
A6	**MLX90640 55° thermal breakout**	waveshare.com/product/mlx90640-d55-thermal-camera.htm	1	$54.99	Alts: Adafruit 4407 $74.95 (out of stock; Newark $73.45, ~3 week lead); The Pi Hut £60 (7 left). 110° "BAA" version has wider view, lower detail.
A7	850 nm IR LEDs, Vishay VSLY5850	vishay.com/docs/83160/vsly5850.pdf	2 (+2 spare)	$2 est.	Datasheet: ±3° half angle, 600 mW/sr at 100 mA. Listed exempt in Vishay's IEC 62471 table. Alt: OSRAM SFH 4725S (940 nm, invisible, less camera sensitivity).
A8	N-ch logic MOSFET, resistors, JST connectors, perfboard	Any	1 kit	$15 est.	
A9	USB-C PD power bank, ≥20,000 mAh, 5 V/5 A PD profile	any certified (UN 38.3)	1	$50 est.	**Must output 5 V/5 A** and not auto-off at low current; check before relying on it (RK-04).
A10	Head strap / helmet mount + belt pouch	Generic	1	$20 est.	Pod 3D-printed (PETG) or off-the-shelf box.
A11	Calibration kit: TMP117 breakout, matte black tape, aluminum block, kettle/ice	Adafruit / hardware store	1	$25 est.	TMP117 ~±0.1 °C class **[UNVERIFIED part/price]**.
A12	Cables, heat-shrink, strain relief, zip ties, spare microSD (64 GB)	Any	1	$20 est.	
A13	**Eye camera**: Arducam OV9281 1 MP mono global-shutter NoIR, CSI, 130° M12 module	arducam.com (OV9281 NoIR MIPI)	1	$41.99	Listed in stock (search listing 2026-10-28); up to 143 fps at 1280×800 RAW8. USB version B0332 $49.99 (120 fps MJPG). Cheaper Pi alternatives $25.90–29.19 not checked for NIR.
A14	M12 lens, ≈25° horizontal FOV (swap for the 130° lens)	any M12 lens vendor	1	$10 est.	Gives ≈0.17 mm/px at 0.5 m. **[UNVERIFIED]** part and price.
A15	Eye IR LED bank: 2× VSLY5850, logic N-FET, resistors	as A7/A8	1	$6 est.	Second bank on GPIO27, own series switch and current limit.
A16	Second Pi 5 camera/display adapter cable (22→15 pin)	pimoroni.com (Pi 5 cables)	1	$5 est.	Price not seen; Pimoroni lists the cables.

#	Item	SKU / link	Qty	Price	Notes and alternatives
A17	I2S MEMS microphone breakout SPH0645LM4H	adafruit.com/product/3421	1	$6.95	Alt: INMP441 modules ($4–7). Pi 5 wiring: GPIO18 BCLK, GPIO19 WS, GPIO20 DIN; I2S driver quirks **[UNVERIFIED]**. Slide switch on 3V3 = hardware mute.
A18	GPS: Adafruit Mini GPS PA1010D (UART + I2C, STEMMA QT)	adafruit.com/product/4415	1	$29.95	In stock when listed. Alt: Ultimate GPS PA1616S $29.95; u-blox NEO-M8N module $17.45 (backordered). Indoors expect no fix.
A19	Haptic: DRV2605L controller + LRA motor	adafruit.com/product/2305	1	$7.95 + $3 est.	Silent alert path (FYI tier).
A20	Buttons (ack, panic), 3V3 slide switch (mic mute), series switch for LED supply	any	1 kit	$6 est.	Panic and ack on GPIO22; mute state read on GPIO23.
	Tier A v2 total			**≈ $461**	v1 $343.99 + $116.84 of new parts = $460.83

Tier B v2: adds second thermal, radar, SDR, audio out, optional secure element (v1 rows B1–B6 unchanged; B7–B10 new).

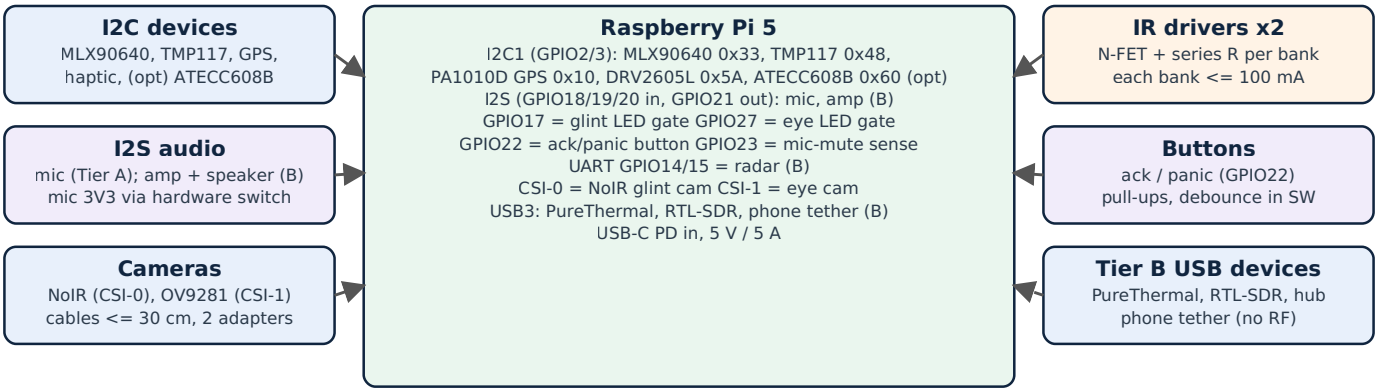
#	Item	SKU / link	Price	Notes and alternatives
B1	FLIR Lepton 3.5 (160×120)	groupgets.com/products/flir-lepton-3-5	$172	86 in stock when checked. Datasheet accuracy ±5 °C typical; glass blocks LWIR.
B2	PureThermal Mini USB (UVC carrier)	groupgets.com/products/purethermal-mini-usb	$114.99	640 in stock when checked. **Confirm whether the Lepton is included before ordering [UNVERIFIED].** Alt: TOPDON TC001 256×192 at $289 (topdon.us); open OpenCV parsers exist.
B3	Seeed MR60BHA2 60 GHz breathing/heartbeat kit	seeedstudio.com (p-5945)	$26.99	Breathing/heartbeat to 1.5 m, presence to 6 m; ESPHome pre-flashed; Pi integration unproven (RK-07).
B4	RTL-SDR Blog **V4L** dongle + antenna	rtl-sdr.com shop	$37.95	V4 is end-of-line ($44.95 / $54.95 with antenna); beware counterfeit marketplace V4s.
B5	Nordic nRF52840 Dongle (BLE sniffer, optional)	Nordic via distributors	$14 est. ($11–16)	Optional second BLE view.
B6	Powered USB 3 hub, short cables	Any	$12 est.	Pi 5 USB budget with PureThermal + SDR.
B7	Adafruit ATECC608 breakout (optional secure element)	adafruit.com/product/4314	$4.95	**Optional, off by default (D-41):** Microchip's CryptoAuthLib license restricts use to Microchip parts (not OSI-open). Key storage without it: LUKS passphrase + phone-held key.
B8	Adafruit MAX98357A I2S amplifier	adafruit.com/product/3006	$5.95	Pi-side speaker for headless mode.
B9	Small 4–8 Ω speaker, 3 W	any	$4 est.	Not worn; belt pack only.
B10	USB-C data cable for phone tether (clip export, no RF)	any	$8 est.	
	Tier B v2 increment		**≈ $401**	with Tier A v2 ≈ $862 (v1 incr. $377.93)

Corrections to v1 parts notes. (1) RTL-SDR Blog V4 tuning range is now verified: 500 kHz–1.766 GHz (vendor datasheet), so it **cannot receive 2.4 GHz BLE/Wi-Fi**; it remains useful only for the optional E-10 memory-emission probe and for sub-GHz sweeps. (2) If the owner later wants IQ-level BLE/Wi-Fi fingerprints (R-3), a 2.4 GHz-capable receive-capable SDR is needed; none is in the BOM and no price was checked. (3) The Pi 5 has two 4-lane MIPI camera ports (confirmed), so the NoIR and eye cameras do not share a port.

Software cost: $0 (all open-source). Apple Developer Program for notarization is not used.

7. Wiring, power, enclosure and safety

Figure 3. Wiring and power plan v2 (3.3 V logic; verify pinouts; GPIO18-21 reserved for I2S)



Power budget v2 (estimates; measure in B-10)

Tier A v2 typical: Pi 5 7.5 W (incl. pupil processing) + 2 cameras 1.7 + thermal 0.1 + glint LEDs 0.15 + eye LEDs 0.2 + GPS 0.1 + haptic 0.1 + mic 0.01 = ~9.9 W typical; peak ~15 W.

Tier B adds PureThermal 0.5, SDR 1.3, radar 0.5, amp + speaker 0.2 avg (1.3 peak): ~12.4 W typical, peak ~19 W; inside the 25 W (5 V/5 A) PD profile.

74 Wh bank at ~85% conversion = ~63 Wh: Tier A ~6.4 h, Tier B ~5.1 h in theory. Requirement: at least 2 h measured, phone link and GPS active.

v1 conflicts fixed: GPIO18 (v1 LED gate) is an I2S clock pin; GPS sits on I2C so the UART stays free for the radar.

Batteries stay on the belt; the phone has its own battery. Pi 5 pin functions follow the Raspberry Pi I2S white paper.

The Pi 5 BCM2712 has ARMv8 crypto extensions (AES, SHA-2, PMULL), so AES-GCM and LUKS are fast.

7.1 Interfaces and constraints (v2)

Interface	Detail	Constraint / check
I^2C1 (GPIO2/3)	MLX90640 0x33, TMP117 0x48 (calibration), PA1010D GPS 0x10, DRV2605L 0x5A, ATECC608B 0x60 (optional)	400 kHz; no address clash; cable ≤30 cm; a second I^2C bus on other pins if GPS traffic disturbs thermal frames **[bench-test]**.
CSI-0 / CSI-1	NoIR Camera Module 3 / OV9281 eye camera (two 4-lane MIPI ports on Pi 5)	Two adapter cables; keep ribbons away from LED wiring.
I^2S (GPIO18/19/20 in, GPIO21 out)	MEMS mic (Tier A), MAX98357A (Tier B)	Pi 5 I^2S pin map per Raspberry Pi white paper; only one I^2S block at a time. **v1's GPIO18 LED gate moved.**
GPIO17 / GPIO27	Glint LED gate / eye LED gate, each N-FET with 100 kΩ pull-down, 100 Ω gate resistor	Hardware series resistor limits each bank to ≤100 mA; LEDs off at boot.
GPIO22 / GPIO23	Ack-panic button / mic-mute sense	Pull-ups, 30 ms debounce; mute switch cuts mic 3V3 in hardware.
UART GPIO14/15 or USB	MR60BHA2 radar (Tier B)	Protocol to confirm (RK-07).
USB 3.0	PureThermal, RTL-SDR, phone tether	Powered hub if undervoltage (`vcgencmd get_throttled`).
BLE / Wi-Fi	Pi 5 onboard; phone BLE	BLE scan with bleak; AP beacons (incl. TSF) from managed-mode `iw scan`; scans windowed so the phone link stays up.

7.2 Power and mounting

Budget and runtime are in Figure 3 (Tier A v2 ≈9.9 W typical/15 W peak; Tier B ≈12.4 W/19 W; ≈6.4 h / 5.1 h theoretical on a 74 Wh bank). A 5 V/3 A supply is not enough: use a certified 25 W (5 V/5 A) PD bank. **Head pod target ≤140 g** (v1: ≤120 g; the eye camera, LEDs, GPS and haptic add weight; recheck in B-11): two cameras, thermal board, four LEDs, haptic, mic, button, GPS on a short lead if it is better on the belt. Cable bundle ≤50 cm with a breakaway connector. Two-part PETG case; LED windows open (no lens, no concentrating diffuser); no cover plate over the thermal sensor; a front label "IR emitter".

7.3 IR LED eye-safety design rules (IEC 62471; standard cited, product not certified)

IEC 62471 classifies lamps and LEDs; for 780–1400 nm IR, the key limits at the standard 200 mm distance are: cornea/lens irradiance ≤100 W/m² for 1000 s (**Exempt**), ≤570 W/m² for 100 s (RG1), ≤3200 W/m² for 10 s (RG2). The retinal weak-visual-stimulus limit is $L_{IR} \leq 6000/\alpha$ W·m^{-2}·sr^{-1} for 1000 s (Exempt). The standard is not itself law but is the basis of regional rules (OSRAM application note); the classification of the final product is the builder's responsibility.

Back-of-envelope for this design (assumption: datasheet values; not independently measured). One VSLY5850 at 100 mA gives 600 mW/sr (900 mW/sr absolute max). At 0.2 m: 0.9 / 0.2^2 = 22.5 W/m² per LED at the absolute-maximum rating; two LEDs ≈ 45 W/m² ≪ 100 W/m² exempt limit. With a 25% duty cycle the time-average is lower still. Vishay lists the part in its exempt table at 850 nm.

Rules we adopt (D-03 default):

1. Only parts that the maker lists as IEC 62471 exempt; **no lasers**; **no lenses, collimators, magnifiers or reflectors** in front of the LEDs.
2. Max 2 LEDs, fixed series resistor for ≤100 mA, software duty ≤25%, a pulse ≤10 ms, and a hardware current check in B-10.
3. Fault rule: if the software watchdog stops, the MOSFET gate pull-down turns the LEDs off; a **physical switch in series with the LED supply** lets the wearer cut them instantly.
4. Aimed at objects by default. Aiming at a consenting person's face is allowed in E-01 only at ≥0.5 m, ≤1 s at a time, with explicit consent, and never at children or pets.
5. If this ever becomes a product or is used on strangers, an accredited IEC 62471 test is required first.

v2 addition: eye illuminator (M-01). A second bank of two VSLY5850 at ≤100 mA, pulsed in sync with 120 fps exposures (≤50% duty, pulse ≤4 ms). Worst case at the closest allowed subject distance of 0.3 m: (0.9 W/sr at the absolute-maximum rating) / 0.3^2 = 10 W/m² per LED; four LEDs on at once ≈ 40 W/m², still below the 100 W/m² exempt limit (calc, datasheet values, not measured). Rules: both banks sit behind the same series switch; subject distance ≥0.3 m for eye work (≥0.5 m for glint work as in v1); LED code is random but bounded by the duty caps; the PLR visible stimulus (lab only) is a separate white LED, ≤1 s, dim start, photometer reading logged, and the visible-light (blue-light) limit of IEC 62471 must be checked against that LED's datasheet **[UNVERIFIED, not done]**. Children and pets are excluded.

7.4 Battery and thermal safety

- Use a commercially certified power bank (UN 38.3 listed), never a bare LiPo cell wired by hand. Inspect for swelling or damage before each session; stop use if it is hot, swollen or smells. OSHA guidance on small and wearable lithium devices applies: a failed cell near the head causes burns or fire.
- Keep the battery on the belt, away from the head and neck; no charging unattended or in the bag; a metal tin or LiPo bag for storage.
- Surface-temperature limit: rig parts at ≤45 °C measured by the thermal camera (E-09). Pi 5 gets an active cooler.

8. Software architecture, licenses and repo layout

Stack. Pi: Raspberry Pi OS 64-bit, Python 3.11+, one asyncio process, a thread per adapter, SQLite, NumPy `.npy`. Phone: Kotlin, minSdk 29, CameraX, SQLCipher, BiometricPrompt, Mapsforge, platform TTS. Shared: one JSON schema for samples, scores, events and the guard vocabulary; the golden dataset checks that Python and Kotlin scores agree within 0.02 on shared features (E-27).

License policy (R-05, unchanged). Allowed: MIT, BSD-2/3, Apache-2.0, ISC, PSF, MPL-2.0, LGPL (dynamic or with exception). GPL/LGPL *tools* only as separate executables. Denied: AGPL, CC-BY-NC, "research only", SSPL/BUSL, any proprietary or non-commercial library, model or dataset.

Component (module)	Purpose	License (checked 2026-10-28 unless noted)	Verdict
bleak (M-03)	BLE scan	MIT	Allowed
OpenCV + Haar eye cascade (M-01, M-07)	pupil, markers	Apache-2.0 (cascade file **[UNVERIFIED]**)	Allowed
scikit-learn, NumPy, SciPy	fusion	BSD-3	Allowed
Tink; BoringSSL inside it (M-04)	Android crypto	Apache-2.0 (BoringSSL: Apache-2.0 with OpenSSL-era BSD parts)	Allowed
libsodium / PyNaCl (M-04)	Pi crypto, Ed25519, constant-time compare	ISC / Apache-2.0 **[PyNaCl license from general knowledge]**	Allowed
noise-java (Android), `noiseprotocol` (Pi)	Noise_XX channel	MIT, MIT	Allowed
SQLCipher Community / sqlcipher-android	encrypted DB	BSD-3 (notice must ship in the app)	Allowed
AndroidX Biometric, CameraX, TFLite	gate, camera, ML	Apache-2.0	Allowed
Mapsforge (M-05)	offline vector map	LGPL-3.0 with §4(d)–(e) waived for apps	Allowed
MapLibre Native Android	alt. map renderer	BSD-2-Clause	Allowed (alt.)
SpeechBrain ECAPA-TDNN weights (M-02)	speaker embedding	Apache-2.0 model card; trained on VoxCeleb (data license **[UNVERIFIED]**)	Lab use; verify data terms
AASIST (M-02)	replay/TTS detection	MIT code; trained on ASVspoof 2019 LA (ODC-By)	Allowed with attribution
Flite (M-06)	Pi TTS	BSD-style	Allowed
sherpa-onnx (M-06)	optional TTS	Apache-2.0; each Piper-VITS voice has its own license	Allowed only per voice
Piper (piper1-gpl), eSpeak NG	TTS	GPL-3.0	**Denied** (default build)
Microchip CryptoAuthLib (ATECC608B)	secure element	proprietary-style, Microchip parts only	**Denied** by default (D-41)

Component (module)	Purpose	License (checked 2026-10-28 unless noted)	Verdict
Google ML Kit, Play Services, Firebase	—	proprietary	**Denied**
cryptsetup/LUKS, `iw`, `rtl_power`	tools	GPL	Separate executables only

```
# asb-rig: standalone repo, Apache-2.0, standalone
LICENSE NOTICE README SECURITY.md
docs/       spec  consent_form  biometric_release  legal_checklist  eye_safety_worksheet
hardware/   pod_cad/  wiring.md  bom.csv
schema/     sample.json score.json event.json vocabulary.json   (shared by Python and Kotlin)
asb/adapters/  thermal_mlx90640 thermal_uvc camera_noir camera_eye ir_led ble_scan wifi_beacons
               mmwave_mr60 imu mic_ptt gps_i2c button_haptic sdr_probe   (+ replay_* and sim_*)
asb/features/  glint thermal_roi pupil blink glint_code rppg voice_embed rf_stats rf_skew baseline
asb/scores/    one module per score (value 0..1, confidence, health, evidence)
asb/fusion/    stack calibrate report          asb/guard/     vocabulary privacy consent retention phrases
asb/security/  keys noise_channel hashchain panic wipe          asb/store/  schema.sql db export sightings sweeps
asb/           cli ui headless/ (replay, virtual clock, stubs)
android/app/   Kotlin: adapters gate sighting map tts sweep fallback
golden/        manifest.json README   (data lives in an encrypted archive, git-ignored)
scripts/       build_all.sh verify_build.sh license_audit.sh run_experiment.py cal_thermal.py cal_align.py
tests/         guard consent hash led_dutycap schema license adapter_contract hashchain crypto headless golden
```

Guard rules (enforced + unit-tested), extended. (1) Regex blocklist for "robot", "android", "fake", "not human", "impostor", "synthetic person", "not alive", "dead", "replica" in any live string about a subject. (2) The output schema has no class field. (3) A human-subject session requires a consent ID. (4) Full-frame NoIR images saved only for non-person objects; humans: 64×64 glint crops, thermal arrays and derived traces only. (5) **No audio is saved**: the mic adapter yields embeddings and spoof scores in RAM; raw audio exists only for the length of one prompt and is never written. (6) **Voice output uses only whitelisted phrase templates.** (7) Refuse to run if monitor mode or a banned radio is found.

9. Data schema (v1 tables kept; additions below)

SQLite (phone: SQLCipher), UTC ISO-8601 plus monotonic ns. v1 tables `session`, `consent`, `device`, `calibration`, `run`, `thermal_frame`, `nir_frame`, `rf_observation`, `radar_sample`, `feature`, `signature_event`, `fusion_result`, `lab_label`, `audit_log` are unchanged (`signature_event.code` now A1–A11; `consent` gains `biometric_release`, `voice_enroll`, `eye_enroll`; `session` gains `site_code`, `gps_policy`).

New table	Key columns	Purpose / privacy note
`eye_trace`	run_id, t_mono_ns, pupil_mm, blink, glint_n, glint_offset_px, glint_code_ok, quality	Derived numbers only; no eye images for humans (non-person props may keep frames).
`voice_event`	run_id, t_mono_ns, prompt_id, spoof_score, embed_sim (nullable), enrolled (bool)	No audio. `embed_sim` only if the speaker enrolled; embeddings are never stored for non-enrolled speakers.
`enrollment`	subject_code, kind (voice/eye), embedding_enc, created_utc, delete_after_utc	AES-GCM encrypted; written release required; deleted on withdrawal.
`emitter_baseline`	site_code, emitter_key, features_json, n_windows, built_utc	Fixed emitters only; `emitter_key` = HMAC of address or BSSID with a per-site key.
`emitter_drift`	emitter_key, t_utc, changed_features, windows	Source of A9; observation only.
`site / anchor`	site_code, name, anchors (marker id, pose), baseline_ids	Per-location baselines.
`sighting`	sighting_id, t_utc, codes, scores, verdict, lat, lon, fix_quality, crop_path, chain_idx	Phone-encrypted; verdict ∈ confirmed/benign/unresolved.
`sweep`	sweep_id, site_code, start/end, counts_json, new_since_last_json, outcome	Timeline source.
`golden_clip`	clip_id, split (train/dev/test/growth/quarantine), label, sha256, source_sighting, consent_ok, manifest_version	The frozen test split hash is recorded in the decision log.
`chain`	idx, h_prev, h_this, anchor_sig	Tamper-evident log; see M-04.

Retention (R-07 extended). 30 days default for sessions, traces, sightings and sweeps unless the owner marks a sighting "keep"; enrollments are deleted on withdrawal or at the session-set end; golden clips from volunteers contain derived features only and are deleted (manifest version bump) on withdrawal; nightly job deletes expired rows and blobs and logs an `audit_log` entry; encryption keys destroyed by the wipe in M-04 make all copies unreadable.

10. Build and calibration procedure (for a later, approved build)

Effort estimate: three weekends for Tier A v2, one more for Tier B, one for the Android app and security. Record every result in `docs/build_log.md`.

10.0 Build steps (nothing done yet; stop at any failed check)

Step	Action	Check (pass condition)
B-00	**Approvals.** the owner approves budget (D-01), repo (D-02), human-subject policy (D-04).	Written OK in the decision log
B-01	Order parts only from Section 5; confirm stock and total at checkout; keep receipts.	Order total ≤ approved budget
B-02	Inspect parts on arrival; photograph serials; note firmware versions.	No damage; serials logged in `device` table
B-03	Flash Raspberry Pi OS 64-bit to microSD; enable I²C, I²S, both cameras, SSH keys only; create the LUKS2 data volume with a typed passphrase; install `git`, Python.	Boots; `i2cdetect -y 1` runs
B-04	Bench-wire the MLX90640 and TMP117 on I²C1 (≤30 cm cable).	`i2cdetect` shows 0x33 and the TMP117 address; frames read at ≥2 Hz
B-05	Connect NoIR camera; capture test frames.	`rpicam-hello` or picamera2 preview works; image not frozen
B-06	Build both IR LED drivers (glint GPIO17, eye GPIO27) on perfboard **with the LEDs disconnected**: verify gate pull-down and current with a resistor load.	Multimeter: 0 mA with GPIO low; ≤100 mA with GPIO high; no current at boot
B-07	Connect LEDs; verify duty cap in software (`test_led_dutycap.py`) and by scope or logged timestamps.	Duty ≤25%, pulse ≤10 ms
B-08	Thermal calibration (see 9.1).	Residual RMS ≤1.0 °C (MLX) at 25–45 °C
B-09	NoIR ↔ thermal alignment (see 9.2).	Reprojection error ≤2 thermal pixels
B-10	Power test: run a 30-min soak with all Tier A sensors; log `vcgencmd get_throttled`, battery voltage if available, surface temps.	No undervoltage flag; surface ≤45 °C; projected runtime ≥2 h
B-11	Build the head pod; weigh it; strain-relieve; test breakaway connector.	Pod ≤120 g; connector releases at light tug
B-12	Run the license audit: `scripts/license_audit.sh`.	No AGPL / non-commercial packages; GPL only as separate executables
B-13	Run the unit tests (guard, consent, hashing, LED cap, schema, adapter contract, hash chain, crypto lint, golden replay).	All pass
B-14	Tier B only: add PureThermal + Lepton, MR60BHA2, RTL-SDR; repeat B-04, B-10 for each.	Each device detected; USB power OK
B-15	Dry run in an empty room (no people), 10 min, all sensors.	Zero crashes; clock skew ≤100 ms; files closed cleanly
B-16	Mount the OV9281 eye camera on CSI-1 with the 25° M12 lens; focus at 0.5 m; print an artificial-pupil target (3, 4, 6 mm discs).	120 fps capture; 4 mm disc ≥20 px at 0.5 m; pupil fit error ≤0.5 mm
B-17	Wire the eye LED bank (LEDs off), then connect; log LED current for both banks, 4 LEDs on together.	Each bank ≤100 mA; combined duty caps hold; series switch cuts both
B-18	Wire mic + hardware mute switch; implement push-to-talk gating.	With the switch off, 0 non-zero samples; with it on, audio only while PTT held; no file written
B-19	Wire GPS (I²C 0x10) and haptic (0x5A); parse NMEA; trigger the three haptic patterns.	Addresses detected; fix or explicit "no fix"; patterns distinguishable
B-20	Pair pod and phone by QR; run the Noise_XX session; force a rekey; replay and bit-flip packets.	Handshake OK; rekey ≤1,000 msgs/10 min; replays and flips rejected
B-21	Create keys: LUKS2 passphrase, Keystore AES-256-GCM key (StrongBox if present), HMAC log key; rehearse crypto-erase on a **scratch** volume only.	Volume unreadable after erase; no plaintext key on disk
B-22	Run `scripts/build_all.sh` on the Mac (or the Linux box for the APK).	Clean tree; APK + macOS artifact carry the same commit hash and SHA-256 list
B-23	Build an offline Mapsforge map for the site region from an OSM extract; test in airplane mode.	Map renders offline; attribution visible; zero network calls
B-24	Record the Phase 0 clips into golden dataset v0; write the manifest; at G0 record the frozen test-split hash.	`verify_golden` passes; hash logged
B-25	Run adapter contract tests for every real, replay and sim adapter.	100% pass; crash isolation shown

10.1 Thermal calibration

1. Warm up the sensor for 10 min (thermal sensors drift while warming up).
2. Target: a flat aluminum block with matte black tape (emissivity ≈0.95 **assumption**), held at a known temperature measured by the TMP117 taped to its surface. Use three reference points: ice-water-cooled block (≈5–10 °C), room (≈22 °C), warm-water-heated block (≈37–45 °C). Never exceed 45 °C.
3. Distance 0.5 m, normal incidence. Fit a two-point linear correction per sensor; store in `calibration`.
4. Verify at one held-out temperature. Record the residual; if >1.0 °C (MLX) the rig does **not** go on to human tests.
5. Reminder: this is not fever screening. FDA guidance for thermography as a medical triage device expects ≤±0.5 °C with a blackbody reference; this rig makes no such claim and never reports body temperature as health data.

10.2 Alignment and timing

- Heat a printed checkerboard pattern backed with a warm pad (cutouts show as thermal contrast) and photograph it with both cameras; compute a homography with OpenCV; log the error.

- Clock: all records use `time.monotonic_ns()`; wall time from the Pi RTC with NTP when online. Max inter-sensor skew target ≤100 ms; BLE and Wi-Fi scans are windowed at 1 s and 10 s and are not expected to be aligned tighter.

11. Test plan, pass/fail criteria and go/no-go gates

Pre-registration rule. The thresholds below are frozen when the owner approves this spec. Any change after data collection starts must be logged in the decision log with the reason and **cannot be used to declare a pass retroactively**. Every experiment is run and logged even when it fails.

Subjects and controls (v1 list plus v2 additions). v1: (a) volunteers with written consent, **the owner first** (D-38); (b) mannequin unheated; (c) mannequin warmed to ≈33–35 °C; (d) powered toy or hobby robot; (e) silicone prop mask or glove on a stand; (f) phones, earbuds, watch; (g) hot drink, laptop, heater; (h) fan or spinning chair; (i) silent object. **v2 adds:** (j) printed face photo and a tablet playing a close-up video of a face (replay attacks); (k) a glass or acrylic craft eye / mannequin with glass eyes (prosthesis-style eye prop); (l) a loudspeaker playing a recorded and a TTS voice (voice replay attacks); (m) a volunteer wearing the silicone prop mask or half-mask with eye holes (report-only: real eyes, pulse, voice remain); (n) 6 owned fixed emitters (ESP32/nRF beacons, own AP, smart plug) for fingerprints; (o) planted objects for baselines (hand warmer, powered USB camera, powered beacon, phone in airplane mode, cardboard, mirror). Props are owned, borrowed or bought only after approval (D-01).

ID	Experiment	Design	Pass criterion (frozen)
E-01	Lens glint	6 camera devices (2 phones, webcam, action cam, laptop cam, a pen/USB spy-style camera if available) and 6 non-camera glints (eyeglasses, watch, jewelry, pen clip, glass bottle, metal faucet); distances 0.5/1/2/3 m; angles 0/10/20/30° off axis; IR-on vs IR-off frame pairs; 3 consenting volunteers' eyes at ≥0.5 m	Camera detection ≥90% at ≤1 m and ≤10° (n ≥ 60 trials); at ≥2 m **report only**. Non-camera false candidates: report; on/off differencing must remove ≥50% of them vs a single-frame threshold.
E-02	Active IR emitters	Known emitters (TV remote 940 nm, phone face-unlock, security-camera IR ring if available) vs negatives, LED off, 1 m lit and 2 m dark	Detect ≥90% of positives; false alarm ≤5% of frames on negatives
E-03	Thermal accuracy	Reference block at 25/33/37/45 °C, 0.5 m; both thermal sensors	Median absolute error after calibration: MLX ≤2.0 °C (target 1.0); Lepton ≤3.0 °C; drift ≤1.0 °C over 30 min after 10 min warm-up
E-04	Thermal separability	Face/hand ROI of 5 humans vs the 4 non-human controls and distractors, at 0.5/1/2 m, rooms at 20–24 °C, plus one cold-hands trial	Report ROC. **Gate G1 input:** thermal-only AUC ≥0.75 for human vs *unheated* mannequin + toy robot + object controls. Human vs *heated* mannequin is **report-only** (expected to be low).
E-05	BLE / Wi-Fi scanning	Known advertisers (phone, watch, earbuds, ESP32 beacon at 0.1/1/10 s intervals) at 1/3/5/10 m; airplane-mode phone and silent object as negatives; observe MAC rotation for 30 min	Detect ≥95% of ≤1 s advertisers within 10 s at ≤5 m; **0 detections** from silent negatives; confirm identifiers rotate (expected) and report what can/can't be linked; nearest-of-3 by RSSI correct ≥70%
E-06	mmWave breathing (Tier B)	Seated still humans at 0.5/1.0/1.5 m; negatives: mannequin, toy robot, oscillating fan	Breathing detected ≥80% within 15 s at ≤1 m; fan false-positive ≤20%. Heart rate is **not** a gate (literature shows weak accuracy).
E-07	End-to-end fusion	≥30 runs per class (human_volunteer, mannequin_unheated, mannequin_heated, toy_robot_powered, object_control); leave-one-subject-out CV; threshold set at human FPR ≤5%	**G1: fused AUC ≥0.80** (human vs unheated mannequin + toy robot + object); report TPR at 5% FPR; heated mannequin result reported separately; every ≥1 sensor failure logged
E-08	Guard and privacy tests	Unit/integration tests	100% pass: no banned strings in any output; no session without consent; no raw MAC stored; retention job deletes test data; full frames not saved when `contains_person=1`; no audio on disk; guard also checks voice phrases
E-09	Safety and endurance	2-h run, 25% LED duty; measure LED current; surface temps with the thermal sensor; pod weight	LED ≤100 mA; surfaces ≤45 °C; runtime ≥2 h; pod ≤140 g; no undervoltage
E-10	Memory-EM probe (optional, exploratory)	RTL-SDR V4L near a known camera while lights toggle, looking for activity in the 24–1766 MHz tuning range (DDR clocks near 800–850 MHz appear in the paper)	**No gate.** Success = reproducible burst correlated with the light toggle at ≥0.5 m. Paper used a USRP B210; a cheap dongle may not work.
E-11	Pupil measurement	Artificial pupil targets (3, 4, 6, 8 mm) on a mannequin face and 5 volunteers at 0.3/0.5/0.75/1.0 m; 120 fps; reference = ruler/target	**Median absolute error ≤0.5 mm at ≤0.5 m; eye-visible frames with a valid pupil ≥90%; 1 m report-only**
E-12	Blink and saccade detection	10 min of video per volunteer at 60 and 120 fps; stimuli at ±5° and ±10°; blinks and saccades hand-annotated; micro-saccade (<1°) estimator run offline	**Blink F1 ≥0.85 at 60 fps; saccades ≥5° recall ≥0.80 at 120 fps; micro-saccades report-only (expected: not detectable beyond noise)**
E-13	Pupil light step (lab only)	Written consent, photometer reading logged, dim start, ≤1 s step, 0.5 m; targets: volunteers, artificial pupils, glass-eye prop	**Volunteers: constriction ≥10% in ≥90% of valid trials; artificial/glass pupils ≤3%; failure rate reported (literature: up to 30%). Not a G1 input**
E-14	Corneal glint geometry and code	random on/off code across the two eye LEDs; volunteers' eyes, glass-eye prop, camera lenses (phones, webcam), printed photo, tablet video; 0.3–1 m	**AUC ≥0.85 volunteer eyes vs {photo, tablet, camera lens, mannequin eye}; code-follow ≥95% on volunteers; glass-eye prop and masked volunteer report-only**

ID	Experiment	Design	Pass criterion (frozen)
E-15	Liveness stack	Volunteers (live) vs attack set {photo, tablet video, mannequin + speaker voice, TTS speaker, toy robot, heated mannequin}; families: optical/thermal, vital, eye, voice; LOSO; ≥20 runs per attack class	**Stack AUC ≥0.90 on the attack set; ≥2-family rule never alerts on a single family; masked-volunteer result report-only; per-family ROC reported**
E-16	Voice liveness and enrollment	≥5 volunteers × 5 sessions; enrolled ECAPA embedding vs impostor volunteers; AASIST on replay (phone, Bluetooth speaker) and TTS; fixed digit prompts; push-to-talk	**Verification EER ≤5% on volunteers (small n: report CI); spoof detection ≥80% at human false-reject ≤10%; no audio file ever written (disk scan); voice never alone raises an alert**
E-17	RF baseline, drift, device type	6 fixed emitters + 3 APs for 24 h baseline; then change one emitter's interval, TX power, payload or firmware; AP skew over 3 days; 12 owned devices for type labels	**Drift detected ≤5 min in ≥90% of 20 injected changes; false drift ≤1 per 24 h static; same-AP skew repeatable within 3σ and distinct APs separable; type accuracy ≥90% with 'unknown' allowed; no IQ fingerprints (hardware cannot)**
E-18	Security tests	Automated + manual per M-04	**100% of: AES-GCM nonce never reused (property test); canary strings absent from disk images; hash-chain verify fails on 1,000 random edits/deletes/reorders and on truncation past the anchor; Noise replay/bit-flip rejected; rekey ≤1,000 msgs/10 min; soft panic ≤1 s; hard panic leaves volume unmountable; 10 failed unlocks trigger erase; constant-time lint clean (no == on secrets)**
E-19	Sighting log and offline map	20 sightings outdoors and indoors; airplane mode; manifest/traffic check; DB opened without key	**GPS stamped ≥95% with a fix; indoor no-fix recorded, never guessed; DB unreadable without key; map renders offline; 0 network requests and no INTERNET permission in the offline build; OSM attribution visible; exports rounded to 0.01°**
E-20	Voice output tiers	50 events per tier; speech start latency; barge-in over a running Attention phrase; airplane mode	**Attention start ≤1.5 s p95; Confirmed barge-in cuts speech ≤300 ms p95; ≤1 Attention per 10 s; 100% of phrases from whitelist; 0 banned words; offline voice only (no network voice)**
E-21	Baselines and new-object alert	3 rooms × 3 baseline passes; 12 planted objects and 12 unplanted control passes; anchors	**New powered emitter at an anchor flagged ≥90% within 2 min; false A10 ≤1 per room-pass on unchanged rooms; thermal blob (≥+5 °C, ≥4 px, ≤1.5 m) ≥80%; passive objects (cardboard, mirror) report-only**
E-22	Thermal grayscale and overlay	Fresh-install and UI tests; readout vs data	**Default is grayscale in 100% of fresh installs; toggle persists; legend and 'not a medical measurement' tag always shown; displayed value equals stored value within 0.1 °C**
E-23	Headless run and golden suite	`asb --headless --replay` on the golden dataset, Linux, no hardware, no network; 3 repeats	**0 failures; guard output byte-identical across repeats; float scores within 1e-6; LED stub asserts never driven; suite ≤2 min laptop, ≤10 min Pi 5; manifest verification 100%**
E-24	One-commit build	`build_all.sh` on a clean Mac; again on dirty tree; Linux APK build	**Refuses dirty tree; APK and macOS artifact carry identical commit hash, SHA-256 list recorded; APK installs and passes app-gate smoke test; macOS self-test runs headless; Linux APK reported (content-equivalence, not bitwise)**
E-25	Adapter contract	Contract suite for every real/replay/sim adapter; kill one adapter thread; unplug a USB device	**100% pass; others continue and the failed one is flagged; re-open ≤10 s; replay timestamps within 1 ms; queue drop counter reported**
E-26	Feedback loop	20 simulated confirmed and 20 benign sightings; attempt to inject into frozen test	**All become quarantined candidates; promotion needs review; clips with people but no consent ID rejected 100%; frozen test manifest hash unchanged 100%; withdrawal bumps manifest and removes clip**
E-27	Android fallback node	≥2 phones of different make; known cameras at ≤1 m; beacons at ≤5 m; Python-Kotlin replay parity	**Glint candidates ≥70% at ≤1 m (reported separately, lower than rig); BLE beacons ≥90% within 10 s at ≤5 m; node never alerts alone; 0 network; score parity within 0.02; ≥2 h runtime at the chosen scan duty**
E-28	Sweep timeline	5 sweeps of one room, 2 with a planted object	**'New since last sweep' shows the planted object in 100% of those sweeps and nothing in the 3 unchanged ones (≤1 spurious entry overall); chain verifies**
E-29	App gate	Class 3 check on test phones; forced UI-callback bypass; lock timing; screenshot	**Auto-lock ≤60 s in background; DB stays undecryptable when the UI callback is bypassed (crypto-bound key); face unlock offered only if Class 3; FLAG_SECURE blocks screenshot; OS lockout observed after repeated failures**

Run protocol (identical for every trial; `scripts/run_experiment.py` enforces the order).

1. Check consent ID (if a person is in the trial), room temperature/lux, and that calibration is valid (≤7 days old). Log battery and sensor health.
2. Warm-up: rig on for ≥10 min; empty-room baseline for 60 s (no subject).

3. Place the subject or object at the marked distance and angle (tape marks at 0.5/1/2/3 m); subject stays still for 15 s, then moves naturally for 15 s where relevant.
4. Capture: thermal 4 Hz, NoIR with LED off/on pairs (LED duty ≤25%), BLE/Wi-Fi windows, radar (Tier B). Operator does not look at the live anomaly index until the run ends (blinding).
5. Operator then enters the ground-truth class in `lab_label` (never before the run).
6. Re-run the empty-room baseline for 30 s; flag drift >1 °C.
7. Repeat in randomized order (a pre-generated schedule, not operator choice); ≥3 repeats per condition.
8. Same day: export a checksum manifest; run E-08 tests; write a one-paragraph run note, including failures.
9. **v2 modalities:** eye camera at 120 fps with the LED code only for runs with a signed biometric release; mic live only during a spoken prompt (push-to-talk); GPS logged only when a fix exists; the headless replay of the finished run must reproduce the live outputs (E-23).

Golden dataset protocol (M-09, M-12; freezes at G0).

1. **Composition at freeze:** ≥100 clips; ≥15 per class (human_volunteer derived traces, mannequin_unheated, mannequin_heated, toy_robot_powered, object_control, replay_attack, planted_object, benign_false_positive); every clip has `manifest.json` entry: SHA-256, label, source run, site, consent ID or "non-person", license, split.
2. **Splits are subject-disjoint:** train 50%, dev 20%, **frozen test 30%**. The frozen-test manifest hash is written to the decision log at G0 and checked at every build; changing it needs a new decision and voids earlier claims.
3. **Growth set:** new confirmed sightings (M-12) go to quarantine → review → growth (train/dev only). Test never grows silently.
4. **Headless regression:** every code change must reproduce expected outputs (E-23); a tolerance change is a logged decision.
5. **Privacy:** human clips contain derived features only; withdrawal removes the clip and bumps `manifest_version`; the archive is encrypted and kept outside git.

Gate G0 (unchanged). Phase 0: a plain phone camera + hand-held 850 nm LED shows a glint from ≥4 of 6 camera devices at ≤1 m; a borrowed thermal camera separates a human from an unheated mannequin at 0.5 m (else moved to B-08); the owner approves the legal checklist. **New at G0:** golden dataset v0 frozen (B-24).

Gate G1 (v2). Phase 2 (glasses or custom firmware) only if E-01 camera detection, E-03, E-05, E-08, **E-18 (security)** and **E-23 (headless)** pass **and** E-07 reaches AUC ≥0.80. If E-07 fails: publish the negative result and stop. If E-07 passes only against unheated proxies (expected): "we can flag unusual devices and cold objects; a determined imitation such as a heated mannequin defeats it."

Module gates (M-gates). Eye (M-01): E-11 and E-14. Liveness (M-02): E-15 and E-16. Fingerprint (M-03): E-17. Baseline (M-07): E-21 (RF part). Fallback (M-13): E-27. Each module whose gate fails **ships disabled by default**; its score is excluded from fusion (missing-family mask) and the failure is written up. Module gates do not block G1, except security and headless as listed.

12. False-positive handling and operator conduct

1. **Alert wording.** "Anomaly signature(s) observed: A3, A4. Not an identification. Common benign causes: cold hands, coat, phone." Never "robot", "fake", "not human".
2. **Threshold policy.** Operating point is chosen from human controls (FPR ≤5%). Alerts below the threshold are logged but not shown.
3. **Evidence threshold.** A single signature (e.g. A4 alone) is displayed as "FYI" only; the anomaly index needs ≥2 independent sensor families to alert.
4. **Human-first review.** The operator must see the contributing sensors; the UI does not hide uncertainty.
5. **No confrontation, no photos, no posting.** Alerts are for the operator's learning during lab tests; Phase 0–1 has no real-world targets.
6. **Hidden-camera sweep mode.** The only mode meant for real use (the owner's own premises). A1/A2 hits are "candidate camera locations", to be confirmed by physical inspection; nothing is called a camera until seen.
7. **Quality flags.** Cold start (<10 min), direct sun, reflective surfaces, subject moving >0.5 m/s, and low battery reduce confidence automatically.
8. **v2 additions.** Eye and voice signatures (A7, A8, A11) are shown only as "indicator not observed" with the benign list, never as a conclusion; prosthetic eyes, contact lenses, sunglasses, eye drops, medication, a cold and a bad mic all trigger them. Spoken alerts use whitelisted phrases; Confirmed tier needs the operator. The operator may press **benign** on any alert; those clips feed the feedback loop (M-12) as false-positive examples. A fallback phone node is advisory only.

13. Risk register

ID	Risk	L	I	Mitigation	Owner
RK-01	Misuse: tool used to label or confront real people	M	H	Guard + schema without class label; phase limits; the owner decision on any public use	the owner
RK-02	IR LED eye injury	L	H	Exempt parts, no optics, hardware current limit, duty cap, consent rules (7.3)	Builder
RK-03	Li-ion pack failure near head	L	H	Certified pack on belt, inspection, no unattended charge (7.4)	Builder
RK-04	Power bank cannot supply 5 V/5 A or auto-shuts off	M	M	Check PD profile first; fall back to 15 W with powered hub, or bench PSU only	Builder

ID	Risk	L	I	Mitigation	Owner
RK-05	Part stock-outs / price jumps (Pi, MLX90640, SDR V4)	H	M	Alternatives listed; verify stock at checkout; budget +15% buffer	the owner
RK-06	Thermal sensors not accurate enough for the question	H	M	Calibration, report-only thresholds for heated mannequin, honest negative result	Builder
RK-07	MR60BHA2 cannot be read directly by the Pi	M	M	Try UART/ESPHome native API/USB serial; drop to Tier B-lite without radar	Builder
RK-08	Legal exposure from recording people or radio use	M	H	Private test environment, written consent, passive-only RF, no jamming, no monitor mode, legal checklist	the owner
RK-09	Biometric-law exposure (BIPA etc.)	L	H	No face images/templates stored; thermal arrays only; no identification	Builder
RK-10	Data leak (laptop/Pi lost)	M	M	LUKS volume, hashed radio IDs, 30-day deletion, no cloud	Builder
RK-11	Project drifts into surveillance product	L	H	Scope statement, decision gate G1, separate repo, explicit non-goals	the owner
RK-12	Confirmation bias (tuning until it "works")	M	M	Frozen thresholds, leave-one-subject-out, negative results published in the report	Builder
RK-13	License contamination (AGPL/non-commercial)	L	M	Allow/deny lists, automated audit B-12	Builder
RK-14	Halo/Mentra delivery delays or restrictions (Phase 2)	H	L	Not on critical path; spike optional	the owner
RK-15	Wi-Fi/BLE scanning disrupts own connectivity	M	L	Scan windows; separate USB adapter if needed	Builder
RK-16	Subject distress or false accusation during tests	L	M	Consent briefing; "we are testing sensors, not people"; right to withdraw	Operator
RK-17	Eye module fails (micro-saccades not detectable; glint code fooled by glass eyes)	H	L	Documented limits; module ships disabled if E-11/E-14 fail; no G1 dependency	Builder
RK-18	Biometric-law exposure for eye/voice data (BIPA, WA, TX, GDPR)	M	H	the owner-first policy, written release, derived-only data, 30-day retention, no strangers, lawyer review before widening	the owner
RK-19	Voice spoofing or false rejection undermines liveness stack	H	L	Voice is low-weight, never alone; report ROC; disable if E-16 fails	Builder
RK-20	Accidental hard erase or data loss	M	M	Two-step hard panic; wipe rehearsed on scratch volume only; golden archive backed up encrypted before any erase test	Builder
RK-21	Lost Pi with unlocked volume or leaked phone	L	M	LUKS2 + Keystore keys, auto-lock 60 s, crypto-erase, no cloud copy	Builder
RK-22	GPS log reveals the owner's home or routines	M	M	Encrypted store, 0.01° export rounding, sightings only on premises, no continuous track	the owner
RK-23	Map data license mistakes (OSM tile policy, ODbL)	L	M	No tile-server downloads, attribution visible, separate layers	Builder
RK-24	Dependency drift: GPL TTS or proprietary SDK creeps in	M	M	License audit in `build_all.sh`; deny-list includes Piper, eSpeak NG, ML Kit	Builder
RK-25	Golden test split leaks into training	M	H	Frozen hash checked every build; growth set separate	Builder
RK-26	Pin map or power budget wrong at bench time	M	L	B-10 soak, I^2S/GPS bench checks (B-18, B-19), keep v1 fallback pins documented	Builder
RK-27	Phone model lacks StrongBox or Class 3 biometrics	M	L	Software Keystore with credential fallback; recorded on the device sheet; face unlock disabled if not Class 3	the owner
RK-28	Free CI assumptions change (GitHub pricing, postponed self-hosted fee)	M	L	Local script is the only required path; recheck pricing before using any runner	the owner

14. Legal and ethics checklist (general information, not legal advice; review before B-01)

Radio

- ☐ **No jamming or blocking of any signal, ever.** The FCC states it is generally illegal to operate, market, sell or import RF jammers; there are no exemptions for personal, business, school or vehicle use. The rig has no transmitter except the Pi's normal client Wi-Fi/Bluetooth and the IR LEDs.
- ☐ **Passive receive only.** Receiving public advertisements (BLE adverts, AP beacons) only. Do **not** decode or retain contents of other people's communications: the federal Wiretap Act (18 U.S.C. §2511) restricts interception; the §2511(2)(g) exception for radio communications "readily accessible to the general public" does not cover cellular or other common-carrier traffic.
- ☐ **No cellular reception.** 47 U.S.C. §302a(d) and 47 CFR §15.121 bar scanners that can receive cellular bands or be easily modified to; the RTL-SDR must not be used to listen to cellular channels, and the software will refuse such center frequencies.
- ☐ No Wi-Fi monitor mode and no data-frame capture in Phase 0–1 (D-05).

People and privacy

- ☐ Written, informed consent from every human subject; the form lists sensors, what is stored, retention, right to withdraw, and that the rig does **not** identify people.
- ☐ Adult volunteers only; no children, no strangers, no public places in Phase 0–1.
- ☐ No face recognition, no face templates, no stored face images. Thermal arrays at 32×24 or 160×120 are not used to identify anyone.

- ☐ **Audio (v2 changed):** the rig now has a mic. It is push-to-talk with a hardware mute switch, records **no ambient audio**, writes no audio files, and listens only to a consenting speaker reading a prompt. California, Florida, Illinois and Washington are all-party consent states for recording (recordinglaw.com, 2026); Texas is one-party. Federal law (18 U.S.C. §2511) still applies to oral communications.
- ☐ Private property: tests happen in the owner's home/office or with the owner's permission. Malls, stores and public venues are out of scope until a legal review.
- ☐ Hidden-camera sweeps limited to premises the owner controls or has permission to inspect.

Ethics

- ☐ Output never labels a person as a robot; only "anomaly signatures".
- ☐ People with prostheses, implants (pacemakers, insulin pumps), hearing aids, wearables and silicone facial prostheses are expected to trigger signatures; we do not test on them without voluntary, informed consent and never publish identifying details.
- ☐ No medical claims: thermal readings are not health data.
- ☐ Negative results are published in the final report.
- ☐ Any change of scope (public use, strangers, selling, or sharing the tool) needs the owner's written decision and a legal review first.

v2: eye, voice and liveness data as sensitive biometrics (not legal advice; no lawyer consulted)

- ☐ **Illinois BIPA** defines biometric identifiers as retina or iris scan, fingerprint, voiceprint, or scan of hand or face geometry (740 ILCS 14/10); requires notice, written release (electronic signature counts after Public Act 103-0769, effective 2024-08-02), a retention and destruction policy, and bars selling them; damages $1,000 negligent / $5,000 intentional or reckless, now one recovery per person for repeated collection by the same method (and a 7th Circuit ruling in 2026 applied that amendment retroactively). Our pupil/blink/glint traces are not iris templates and we store no iris images, but a voice *enrollment* is a voiceprint; so every enrolled volunteer signs a written biometric release regardless of state. Applies if the owner or a volunteer is in Illinois.
- ☐ **Texas CUBI** (informed consent before capturing a biometric identifier for a commercial purpose) and **Washington RCW 19.375** (voiceprints and other identifiers; notice, consent or opt-out for commercial enrollment) treat voiceprints similarly; **California** separately requires prior express written consent for voiceprint or voice-stress testing. Our use is non-commercial research on volunteers, which may fall outside some statutes; **[UNVERIFIED]** and not relied on.
- ☐ **GDPR Art. 9** (if any EU person is involved): biometric data for unique identification is special-category; a DPIA-style note is kept even for household research. We do not identify anyone, which weakens but does not remove the exposure.
- ☐ **Purpose limitation:** eye/voice data is used only for the lab liveness experiments; never for identifying strangers, never sold or shared, deleted on withdrawal.
- ☐ **GPS and sighting data:** location of the owner's premises is personal data; encrypted, rounded on export, no tracking of other people.
- ☐ **Anti-tamper and wipe:** protects the owner's own data on the owner's own device. Do not use crypto-erase or the panic gesture to destroy records that are subject to a legal hold or an investigation; if a dispute, subpoena or complaint is pending, stop wiping and ask counsel.
- ☐ **Hidden-camera findings:** a "confirmed" sighting means the operator physically found a device on premises the owner controls. Do not remove, tamper with, or publish about a device found elsewhere; contact the property owner or police as appropriate.
- ☐ **Open-source licenses:** keep SQLCipher's BSD notice and the ODC-By attribution for ASVspoof-derived models in the app's licenses screen; show "© OpenStreetMap contributors".

15. Decision log: settled defaults and residual items

the owner's instruction was to ask nothing and settle everything; every row below is a **recorded default** that the owner may overturn later (changes after data collection must be logged and cannot rescue a failed criterion). Only the residual table needs a human.

ID	Topic	Settled default and reason
D-02	Repo and license	New repo `asb-rig`, Apache-2.0, standalone.
D-03	IR policy	Exempt-group parts, ≤2 LEDs per bank (two banks), ≤100 mA, duty caps, no optics, series switch; subject ≥0.3 m (eye) / ≥0.5 m (glint).
D-04	Subjects	Household adults with a written release; no strangers, no children, private premises only.
D-05	RF scope	BLE adverts + AP beacons (incl. TSF) in managed mode; no monitor mode, cellular, data frames, or jamming.
D-05b	Prosthesis control	Silicone prop and glass-eye prop only; no volunteer with a prosthesis for now.
D-06	Halo	Not in the default build (vendor-licensed SDK portion).
D-07	Glasses spike	Skip until G1.
D-08	Real humanoid test	Skip; optionally watch a public demo without recording anyone.
D-09	Platform	Rig + Android operator app; glasses Phase 2 only after G1.
D-10	Thermal	MLX90640 primary, Lepton/UVC Tier B.
D-11	Fusion	Stacked logistic regression over ≤6 family values; LOSO; human FPR ≤5%.
D-12	Storage	LUKS2 (Pi), SQLCipher (phone), 30-day retention, hashed IDs, no cloud.
D-13	Public or stranger use	Never in Phase 0–1.

ID	Topic	Settled default and reason
D-14	Legal review	Self-review now; a lawyer's review before any subject beyond household adults, any public use, or any sale/sharing.
D-15	Eye scope	Near-range (0.3–1 m) pupil, blink, glint-code; micro-saccades report-only; light-step PLR lab-only and off by default.
D-16	Eye hardware	OV9281 mono CSI, 25° M12 lens, two 850 nm LEDs pulsed in sync (Section 6).
D-17	Face handling	Operator-aimed; transient eye-ROI detection in RAM; no templates; no stored face crops.
D-18	Voice	Push-to-talk, hardware mute, no ambient audio, no audio files; embeddings ephemeral unless written enrollment; liveness-consistency purpose only.
D-19	Liveness stack	≥2 families; outputs "indicators not observed"; voice never alone; module ships disabled if its gate fails.
D-20	Fingerprints	Behavioral BLE/AP + AP clock skew on fixed owned emitters; no IQ-level RFFI; drift = changed ≥2 features for ≥3 windows.
D-21	Crypto primitives	AES-256-GCM (Keystore, files) / AES-256-XTS (LUKS2); ChaCha20-Poly1305 and X25519 in Noise; Ed25519 anchors; SHA-256, BLAKE2s; no custom crypto.
D-22	Key rotation	New Noise handshake every session; Rekey ≤1,000 messages or 10 min; static keys 90 days; no double ratchet.
D-23	Offline vs remote wipe	Crypto-erase; triggers: panic, 10 failed unlocks, signed wipe from the paired phone in BLE range; cloud relay OFF; dead-man OFF.
D-24	Tamper-evident log	HMAC hash chain + Ed25519 anchors on the phone every 60 s or 100 records.
D-25	GPS	Stored encrypted at full resolution; exports rounded to 0.01°; no continuous tracking; no fix = "no fix".
D-26	Maps	Mapsforge + OSM extracts; no tile.openstreetmap.org downloads; attribution on screen.
D-27	TTS	Platform TTS with an offline voice (phone), Flite (Pi); Piper and eSpeak NG excluded.
D-28	Voice tiers	FYI = haptic; Attention = whitelisted phrase, 1 per 10 s; Confirmed = barge-in, operator confirmation needed.
D-29	Baselines	RF-first; optical comparison only at ArUco anchors; optical A10 report-only.
D-30	Thermal display	Grayscale 10–45 °C fixed scale default; color overlay is a toggle.
D-31	Headless/golden	Replay adapters, virtual clock, subject-disjoint splits, frozen test hash at G0.
D-32	macOS CI	Spec-only: local `build_all.sh`; no paid Actions; self-hosted free runner optional; unsigned macOS artifact.
D-33	Adapter layer	Python Protocol + Kotlin interface from one JSON schema.
D-34	Feedback loop	Quarantine → review → growth set; frozen test never changes.
D-35	Fallback node	AOSP/Apache only; minSdk 29; advisory only; never alerts alone.
D-36	Sweep timeline	SQLite/SQLCipher, hash-chained, "new since last sweep".
D-37	App gate	BIOMETRIC_STRONG + device credential; crypto-bound; 60 s lock; FLAG_SECURE.
D-38	Rollout order	the owner alone first, then household adults.
D-39	Biometric retention	Raw eye/voice never stored; derived traces 30 days; enrollments until withdrawal or set end.
D-40	GPIO map	Figure 3 (GPIO17/27 LED gates, I2S on 18–21, GPS on I^2C).
D-41	Secure element	Not in default build (CryptoAuthLib is not OSI-open); optional ATECC608B listed in Tier B.
D-42	Phone	the owner's existing Android phone (assumed owned; not bought).
D-43	Pod weight	≤140 g.
D-44	Golden composition	≥100 clips, ≥15 per class, 50/20/30 splits.
D-45	Anti-tamper scope	Protects the owner's data on the owner's device only.

Residual items (need the owner or a fact only the owner has)

ID	Item	Default until answered
D-01	**Purchase approval.** Tier A v2 ≈ $461; Tier B v2 +$401; props +$50–150; Phase 2 glasses +$399–449.	Nothing is bought. Recommend Tier A v2 first, Tier B only after G0.
R-1	the owner's phone model: StrongBox and Class 3 biometric support are unknown.	Software Keystore + device-credential fallback; record at B-22.
R-2	Will `asb-rig` be public or private? Public repos get free hosted macOS minutes; private ones cost $0.062/min.	Private; local build only.
R-3	IQ-level BLE/Wi-Fi fingerprinting would need a 2.4 GHz receive SDR (not budgeted, price unchecked).	Not planned.
R-4	Where the owner and volunteers live (which biometric/recording statutes apply).	Strictest rules everywhere: written release, all-party recording consent.
R-5	Optional user-owned wipe relay endpoint.	OFF.

Appendix A. Sources (all accessed 2026-10-28)

v1 sources 1–25 remain valid (Halo, Mentra, Meta, Raspberry Pi, part vendors, Melexis, FLIR, Seeed, RTL-SDR, Nordic, Vishay/OSRAM/IEC 62471, FDA/AAMI, FCC, 18 U.S.C. §2511, BIPA, Eye of Sauron, bleak, OSHA, mmWave reviews); see the v1 PDF's Appendix A. New in v2 (items marked **(secondary)** are blogs, vendor pages or aggregators):

- **S26.** Eye tracking: Tobii, "Eye tracker sampling frequency" (connect.tobii.com); PMC7880983 (Tobii Pro Spectrum and microsaccades); RemoteEye, Behav. Res. Methods (doi 10.3758/s13428-019-01305-2); PMC7141092 (60 vs 120 Hz mobile saccade detection); JOV 2800208 (limits of iPad eye tracking).
- **S27.** Pupillometry: PupilScreen 2nd generation (Neurosurgery abstract, doi 10.1227/neu.0000000000001880_477); PMC10686294 (at-home smartphone pupillometry); Evaluation of two portable pupillometers (doi 10.2217/cnc-2020-0016, 30% vs 4% failures); Int J Appl Basic Med Res 14(1):23 (2024).
- **S28.** Corneal glint and eye liveness: Y-EYES (ACM, doi 10.1145/3469261.3469408); AIME (doi 10.1145/3458864.3466911); Morales et al., Introduction to Iris PAD (biometrics.eps.uam.es); PAD for iris recognition review, arXiv 1804.00194 (prosthetic eyes); Attack of mechanical replicas, IEEE TIFS (doi 10.1109/tifs.2015.2405345).
- **S29.** RF fingerprints: Evaluating Physical-Layer BLE Location Tracking (IEEE S&P 2022, UCSD); Identifying BLE devices (ACM SenSys 2021, doi 10.1145/3485730.3492880); Robust RFFI for BLE (IEEE WCNC 2025, doi 10.1109/wcnc61545.2025.10978258); TI BLE5-Stack clock accuracy guide (±50/±500 ppm); Jana and Kasera, MobiCom 2008 (doi 10.1145/1409944.1409958).
- **S30.** Voice: ASVspoof 5 (arXiv 2408.08739; Comput. Speech Lang. 2025, doi 10.1016/j.csl.2025.101825); Vulnerability issues in ASV, EURASIP J. Audio Speech Music (2024); SpeechBrain spkrec-ecapa-voxceleb (Hugging Face model card); clovaai/aasist LICENSE (MIT).
- **S31.** Android security: developer.android.com/privacy-and-security/keystore, KeyGenParameterSpec.Builder, security-key-attestation; developer.android.com/identity/sign-in/biometric-auth; source.android.com biometric and attestation pages; OWASP MASTG-KNOW-0001; Android Bluetooth permissions and Wi-Fi scanning docs; Android BLE scan limits (Silicon Labs docs and Stack Overflow **(secondary)**).
- **S32.** Crypto libraries and protocols: google/tink LICENSE (Apache-2.0); BoringSSL LICENSE; noiseprotocol.org (Noise spec); signal.org Double Ratchet specification; rweather/noise-java (MIT); sqlcipher/sqlcipher-android LICENSE (BSD-3).
- **S33.** Maps: OSM Foundation Tile Usage Policy (operations.osmfoundation.org/policies/tiles/); OSMF Licence/Attribution Guidelines and Legal FAQ; mapsforge/mapsforge COPYING (LGPL-3.0 with exceptions); maplibre-native Android LICENSE (BSD-2); OpenAndroMaps license page; organicmaps (Apache-2.0).
- **S34.** TTS: OHF-Voice/piper1-gpl (GPL-3.0); espeak-ng (GPL-3.0); k2-fsa/sherpa-onnx (Apache-2.0) and TTS APK docs; Piper voice licensing discussion (rhasspy/piper #271) **(secondary)**; Flite (BSD) **(secondary listing)**.
- **S35.** CI: GitHub Docs, "Billing and usage" and "Actions runner pricing" ($0.062/min standard macOS private repos; Linux $0.006/min; self-hosted free for now); GitHub "Pricing changes for GitHub Actions" (2026).
- **S36.** Parts: Arducam OV9281 listings (arducam.com, Amazon, RobotShop) **(secondary)**; Adafruit products 4415 (PA1010D), 746, 4279, 3421 (SPH0645), 3006 (MAX98357A), 2305 (DRV2605L), 4314 (ATECC608); Addicore NEO-M8N; Pimoroni Pi 5 camera cables; RTL-SDR Blog V4 datasheet (500 kHz–1.766 GHz); Raspberry Pi 5 product brief and I2S white paper (RP-009699); Microchip CryptoAuthLib license.txt.
- **S37.** Law: Illinois PA 103-0769 (ilga.gov); ABA Business Law Today and Sidley (7th Cir. 2026 retroactivity) **(secondary)**; recordinglaw.com state recording and biometric pages **(secondary)**; Justia 50-state survey **(secondary)**; Chapter 19.375 RCW (app.leg.wa.gov).

Appendix B. Still unverified or deliberately not done

- **Everything v1 listed** (stock, "est." prices, PureThermal contents, MR60BHA2 on a Pi, Mentra stock/IR, Halo details, FCC interception guide text, state hidden-camera laws, IR-LED measurements) still applies.
- **New:** price and part of the M12 25° lens, second Pi 5 camera cable, LRA motor, LED/driver parts, speaker, USB tether, buttons; OV9281 stock (listing only); I^2S mic driver quirks on Pi 5; GPS on I^2C alongside thermal at 400 kHz.
- Licenses read from search summaries, not from the repos' files: Haar eye cascade, PyNaCl, `noiseprotocol` (Python), SpeechBrain/AASIST *training-data* terms (VoxCeleb), Mapsforge map-file terms, Flite voices, CHROM/POS rPPG patent status.
- Photobiological (blue-light) limit for the visible PLR stimulus LED: not calculated.
- All pass thresholds in E-11…E-29 are my proposals, not published benchmarks. Eye, voice and liveness performance against a real humanoid is **unknown**; no study covers it.
- Railway as a Linux build host: not priced or tested. Apple Developer Program fee: not checked.
- State biometric statutes beyond IL, TX, WA, CA were not surveyed; nothing here is legal advice.